

CHAPTER 14

fixed-income products and analysis: yield, duration and convexity



The aim of this Chapter...

...is to introduce the most common contracts of the fixed-income world and to show simple ways for their analysis. The big assumption of this chapter is that interest rates are deterministic. Greater levels of sophistication are needed for pricing more complex fixed-income contracts such as derivatives, but these will be reached in later chapters.

In this Chapter...

- the names and properties of the basic and most important fixed-income products
- the definitions of features commonly found in fixed-income products
- simple ways to analyze the market value of the instruments: yield, duration and convexity
- how to construct yield curves and forward rates

14.1 INTRODUCTION

This chapter is an introduction to some basic instruments and concepts in the world of fixed income, that is, the world of cashflows that are in the simplest cases independent of any stocks, commodities, etc. I will describe the most elementary of fixed-income instruments, the coupon-bearing bond, and show how to determine various properties of such bonds to help in their analysis.

This chapter is self contained, and does not require any knowledge from earlier chapters. A lot of it is also not really necessary reading for anything that follows. The reason for this is that, although the concepts and techniques I describe here are used in practice and are *useful* in practice, it is difficult to present a completely coherent theory for more sophisticated products in this framework.

14.2 SIMPLE FIXED-INCOME CONTRACTS AND FEATURES

14.2.1 The zero-coupon bond

The **zero-coupon bond** is a contract paying a known fixed amount, the **principal**, at some given date in the future, the **maturity** date T . For example, the bond pays \$100 in 10 years' time, see Figure 14.1. We're going to scale this payoff so that in future all principals will be \$1.

This promise of future wealth is worth something now. It cannot have zero or negative value. Furthermore, except in extreme circumstances, the amount you pay initially will be smaller than the amount you receive at maturity.

We discussed the idea of time value of money in Chapter 1. This is clearly relevant here and we will return to this in a moment.

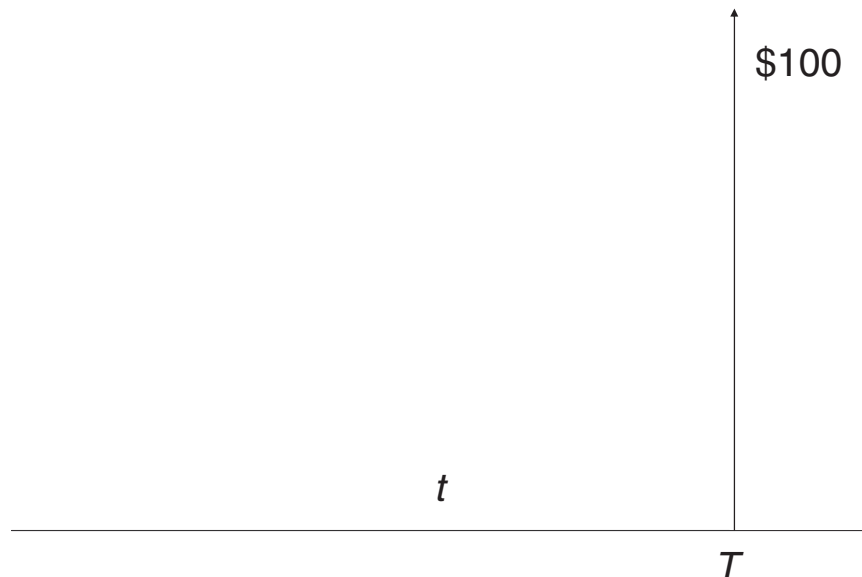


Figure 14.1 The zero-coupon bond.

14.2.2 The coupon-bearing bond

A **coupon-bearing bond** is similar to the above except that as well as paying the principal at maturity, it pays smaller quantities, the coupons, at intervals up to and including the maturity date. See Figure 14.2.

These coupons are usually prespecified fractions of the principal. For example, the bond pays \$1 in 10 years and 2%, i.e. 2 cents, every six months. This would be called a 4% coupon. This bond is clearly more valuable than the bond in the previous example because of the coupon payments. We can think of the coupon-bearing bond as a portfolio of zero-coupon bearing bonds; one zero-coupon bearing bond for each coupon date with a principal being the same as the original bond's coupon, and then a final zero-coupon bond with the same maturity as the original.

Figure 14.3 is an excerpt from *The Wall Street Journal Europe* of 14th April 2005 showing US Treasury bonds, notes and bills. Observe that there are many different 'rates' or coupons, and different maturities. The values of the different bonds will depend on the size of the coupon, the maturity and the market's view of the future behavior of interest rates.

14.2.3 The money market account

Everyone who has a bank account has a **money market account**. This is an account that accumulates interest compounded at a rate that varies from time to time. The rate at which interest accumulates is usually a short-term and unpredictable rate. In the sense that money held in a money market account will grow at an unpredictable rate, such an account is risky when compared with a one-year zero-coupon bond. On the other hand, the money market account can be closed at any time but if the bond is sold before maturity there is no guarantee how much it will be worth at the time of the sale.

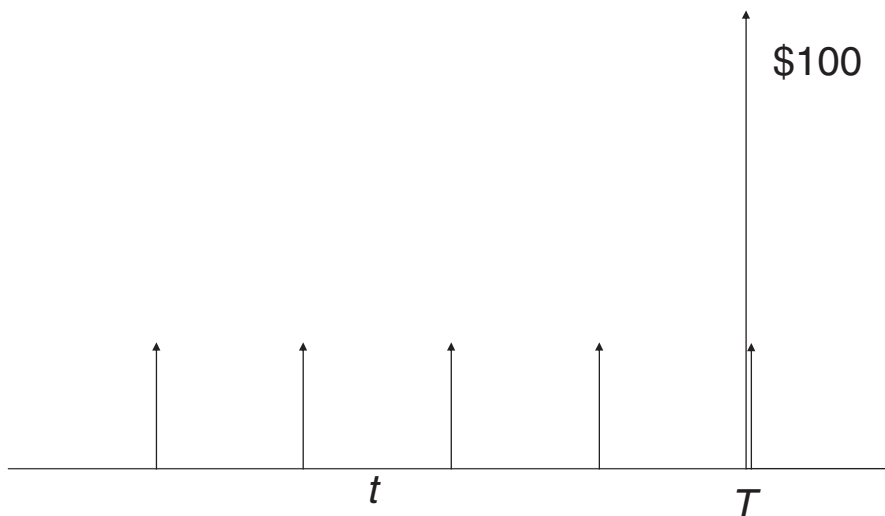


Figure 14.2 The coupon-bearing bond.

Source: Reuters Ltd.																																																																																																																																																																																																																																																																																																																																																																																																																														
TREASURY BONDS, NOTES AND BILLS																																																																																																																																																																																																																																																																																																																																																																																																																														
Wednesday, April 13, 2005																																																																																																																																																																																																																																																																																																																																																																																																																														
Explanatory Notes																																																																																																																																																																																																																																																																																																																																																																																																																														
Representative Over-the-Counter quotation based on transactions of \$1 million or more. Treasury bond, note and bill quotes are as of mid-afternoon. Colons in bid-and-asked quotes represent 32nds; 101:01 means 101 1/32. Net changes in 32nds. n-Treasury note. I-Inflation-Indexed issue. Treasury bill quotes in hundredths, quoted on terms of a rate of discount. Days to maturity calculated from settlement date. All yields are to maturity and based on the asked quote. Latest 13-week and 26-week bills are boldfaced. For bonds callable prior to maturity, yields are computed to the earliest call date for issues quoted above par and to the maturity date for issues below par. *When issued. Source: eSpeed/Cantor Fitzgerald																																																																																																																																																																																																																																																																																																																																																																																																																														
<table border="1"> <thead> <tr> <th>RATE</th><th>MATURITY</th><th>BID</th><th>ASKED</th><th>CHG</th><th>ASK</th><th>YLD</th></tr> </thead> <tbody> <tr> <td>1.625</td><td>Apr 05n</td><td>99:29</td><td>99:30</td><td>---</td><td>2.67</td><td></td></tr> <tr> <td>6.500</td><td>May 05n</td><td>100:10</td><td>100:11</td><td>---</td><td>2.41</td><td></td></tr> <tr> <td>6.750</td><td>May 05n</td><td>100:10</td><td>100:11</td><td>---</td><td>2.48</td><td></td></tr> <tr> <td>12.000</td><td>May 05</td><td>100:25</td><td>100:26</td><td>---</td><td>2.20</td><td></td></tr> <tr> <td>1.250</td><td>May 05n</td><td>99:26</td><td>99:27</td><td>1</td><td>2.45</td><td></td></tr> <tr> <td>1.125</td><td>Jun 05n</td><td>99:21</td><td>99:22</td><td>---</td><td>2.59</td><td></td></tr> <tr> <td>1.500</td><td>Jul 05n</td><td>99:18</td><td>99:19</td><td>---</td><td>2.81</td><td></td></tr> <tr> <td>6.500</td><td>Aug 05n</td><td>101:05</td><td>101:06</td><td>---</td><td>2.89</td><td></td></tr> <tr> <td>10.750</td><td>Aug 05</td><td>102:21</td><td>---</td><td>---</td><td>2.81</td><td></td></tr> <tr> <td>2.000</td><td>Aug 05n</td><td>99:19</td><td>99:20</td><td>---</td><td>2.95</td><td></td></tr> <tr> <td>1.625</td><td>Sep 05n</td><td>99:10</td><td>99:11</td><td>---</td><td>3.06</td><td></td></tr> <tr> <td>1.625</td><td>Oct 05n</td><td>99:05</td><td>99:06</td><td>---</td><td>3.14</td><td></td></tr> <tr> <td>5.750</td><td>Nov 05n</td><td>101:13</td><td>101:14</td><td>---</td><td>3.22</td><td></td></tr> <tr> <td>5.875</td><td>Nov 05n</td><td>101:16</td><td>101:17</td><td>---</td><td>3.21</td><td></td></tr> <tr> <td>1.875</td><td>Nov 05n</td><td>99:05</td><td>99:06</td><td>---</td><td>3.19</td><td></td></tr> <tr> <td>1.875</td><td>Dec 05n</td><td>99:00</td><td>99:01</td><td>---</td><td>3.24</td><td></td></tr> <tr> <td>1.875</td><td>Jan 06n</td><td>98:27</td><td>98:28</td><td>---</td><td>3.29</td><td></td></tr> <tr> <td>5.625</td><td>Feb 06n</td><td>101:29</td><td>101:30</td><td>---</td><td>3.26</td><td></td></tr> <tr> <td>9.375</td><td>Feb 06</td><td>105:01</td><td>105:02</td><td>---</td><td>3.19</td><td></td></tr> <tr> <td>1.625</td><td>Feb 06n</td><td>98:15</td><td>98:16</td><td>---</td><td>3.35</td><td></td></tr> <tr> <td>1.500</td><td>Mar 06n</td><td>98:06</td><td>98:07</td><td>---</td><td>3.38</td><td></td></tr> <tr> <td>2.250</td><td>Apr 06n</td><td>98:25</td><td>98:26</td><td>---</td><td>3.40</td><td></td></tr> <tr> <td>2.000</td><td>May 06n</td><td>98:15</td><td>98:16</td><td>---</td><td>3.40</td><td></td></tr> <tr> <td>4.625</td><td>May 06n</td><td>101:08</td><td>101:09</td><td>---</td><td>3.41</td><td></td></tr> <tr> <td>6.875</td><td>May 06n</td><td>103:20</td><td>103:21</td><td>---</td><td>3.40</td><td></td></tr> <tr> <td>2.500</td><td>May 06n</td><td>98:30</td><td>98:31</td><td>---</td><td>3.42</td><td></td></tr> <tr> <td>2.750</td><td>Jun 06n</td><td>99:04</td><td>99:05</td><td>---</td><td>3.45</td><td></td></tr> <tr> <td>7.000</td><td>Jul 06n</td><td>104:08</td><td>104:09</td><td>---</td><td>3.48</td><td></td></tr> <tr> <td>2.750</td><td>Jul 06n</td><td>99:01</td><td>99:02</td><td>---</td><td>3.48</td><td></td></tr> <tr> <td>2.375</td><td>Aug 06n</td><td>98:16</td><td>98:17</td><td>---</td><td>3.51</td><td></td></tr> <tr> <td>2.375</td><td>Aug 06n</td><td>98:14</td><td>98:15</td><td>---</td><td>3.52</td><td></td></tr> <tr> <td>2.500</td><td>Sep 06n</td><td>98:15</td><td>98:16</td><td>---</td><td>3.55</td><td></td></tr> <tr> <td>6.500</td><td>Oct 06n</td><td>104:08</td><td>104:09</td><td>---</td><td>3.55</td><td></td></tr> <tr> <td>2.500</td><td>Oct 06n</td><td>98:12</td><td>98:13</td><td>---</td><td>3.57</td><td></td></tr> <tr> <td>2.625</td><td>Nov 06n</td><td>98:16</td><td>98:17</td><td>---</td><td>3.57</td><td></td></tr> <tr> <td>3.500</td><td>Nov 06n</td><td>99:27</td><td>99:28</td><td>---</td><td>3.57</td><td></td></tr> <tr> <td>2.875</td><td>Nov 06n</td><td>98:27</td><td>98:28</td><td>---</td><td>3.58</td><td></td></tr> <tr> <td>3.000</td><td>Dec 06n</td><td>98:30</td><td>98:31</td><td>---</td><td>3.61</td><td></td></tr> <tr> <td>3.375</td><td>Jan 07n</td><td>105:00</td><td>105:01</td><td>---</td><td>3.61</td><td></td></tr> <tr> <td>3.125</td><td>Jan 07n</td><td>99:03</td><td>99:04</td><td>---</td><td>3.62</td><td></td></tr> <tr> <td>2.250</td><td>Feb 07n</td><td>97:17</td><td>97:18</td><td>---</td><td>3.62</td><td></td></tr> <tr> <td>6.250</td><td>Feb 07n</td><td>104:22</td><td>104:23</td><td>---</td><td>3.57</td><td></td></tr> <tr> <td>3.375</td><td>Feb 07n</td><td>99:16</td><td>99:17</td><td>---</td><td>3.63</td><td></td></tr> <tr> <td>3.750</td><td>Mar 07n</td><td>100:05</td><td>100:06</td><td>---</td><td>3.65</td><td></td></tr> <tr> <td>6.625</td><td>May 07n</td><td>105:27</td><td>105:28</td><td>---</td><td>3.66</td><td></td></tr> <tr> <td>4.375</td><td>May 07n</td><td>101:12</td><td>101:13</td><td>---</td><td>3.66</td><td></td></tr> <tr> <td>3.125</td><td>May 07n</td><td>98:28</td><td>98:29</td><td>---</td><td>3.67</td><td></td></tr> <tr> <td>2.750</td><td>Aug 07n</td><td>97:27</td><td>97:28</td><td>---</td><td>3.71</td><td></td></tr> <tr> <td>3.250</td><td>Aug 07n</td><td>98:30</td><td>98:31</td><td>---</td><td>3.71</td><td></td></tr> <tr> <td>6.125</td><td>Aug 07n</td><td>105:11</td><td>105:12</td><td>---</td><td>3.70</td><td></td></tr> <tr> <td>3.000</td><td>Nov 07n</td><td>98:05</td><td>98:06</td><td>---</td><td>3.74</td><td></td></tr> <tr> <td>3.625</td><td>Jan 08n</td><td>107:16</td><td>107:17</td><td>---</td><td>3.85</td><td></td></tr> <tr> <td>3.000</td><td>Feb 08n</td><td>97:26</td><td>97:27</td><td>---</td><td>3.80</td><td></td></tr> <tr> <td>5.500</td><td>Feb 08n</td><td>104:18</td><td>104:19</td><td>---</td><td>3.77</td><td></td></tr> <tr> <td>3.375</td><td>Feb 08n</td><td>98:27</td><td>98:28</td><td>---</td><td>3.80</td><td></td></tr> <tr> <td>2.625</td><td>May 08n</td><td>96:15</td><td>96:16</td><td>---</td><td>3.83</td><td></td></tr> </tbody> </table>		RATE	MATURITY	BID	ASKED	CHG	ASK	YLD	1.625	Apr 05n	99:29	99:30	---	2.67		6.500	May 05n	100:10	100:11	---	2.41		6.750	May 05n	100:10	100:11	---	2.48		12.000	May 05	100:25	100:26	---	2.20		1.250	May 05n	99:26	99:27	1	2.45		1.125	Jun 05n	99:21	99:22	---	2.59		1.500	Jul 05n	99:18	99:19	---	2.81		6.500	Aug 05n	101:05	101:06	---	2.89		10.750	Aug 05	102:21	---	---	2.81		2.000	Aug 05n	99:19	99:20	---	2.95		1.625	Sep 05n	99:10	99:11	---	3.06		1.625	Oct 05n	99:05	99:06	---	3.14		5.750	Nov 05n	101:13	101:14	---	3.22		5.875	Nov 05n	101:16	101:17	---	3.21		1.875	Nov 05n	99:05	99:06	---	3.19		1.875	Dec 05n	99:00	99:01	---	3.24		1.875	Jan 06n	98:27	98:28	---	3.29		5.625	Feb 06n	101:29	101:30	---	3.26		9.375	Feb 06	105:01	105:02	---	3.19		1.625	Feb 06n	98:15	98:16	---	3.35		1.500	Mar 06n	98:06	98:07	---	3.38		2.250	Apr 06n	98:25	98:26	---	3.40		2.000	May 06n	98:15	98:16	---	3.40		4.625	May 06n	101:08	101:09	---	3.41		6.875	May 06n	103:20	103:21	---	3.40		2.500	May 06n	98:30	98:31	---	3.42		2.750	Jun 06n	99:04	99:05	---	3.45		7.000	Jul 06n	104:08	104:09	---	3.48		2.750	Jul 06n	99:01	99:02	---	3.48		2.375	Aug 06n	98:16	98:17	---	3.51		2.375	Aug 06n	98:14	98:15	---	3.52		2.500	Sep 06n	98:15	98:16	---	3.55		6.500	Oct 06n	104:08	104:09	---	3.55		2.500	Oct 06n	98:12	98:13	---	3.57		2.625	Nov 06n	98:16	98:17	---	3.57		3.500	Nov 06n	99:27	99:28	---	3.57		2.875	Nov 06n	98:27	98:28	---	3.58		3.000	Dec 06n	98:30	98:31	---	3.61		3.375	Jan 07n	105:00	105:01	---	3.61		3.125	Jan 07n	99:03	99:04	---	3.62		2.250	Feb 07n	97:17	97:18	---	3.62		6.250	Feb 07n	104:22	104:23	---	3.57		3.375	Feb 07n	99:16	99:17	---	3.63		3.750	Mar 07n	100:05	100:06	---	3.65		6.625	May 07n	105:27	105:28	---	3.66		4.375	May 07n	101:12	101:13	---	3.66		3.125	May 07n	98:28	98:29	---	3.67		2.750	Aug 07n	97:27	97:28	---	3.71		3.250	Aug 07n	98:30	98:31	---	3.71		6.125	Aug 07n	105:11	105:12	---	3.70		3.000	Nov 07n	98:05	98:06	---	3.74		3.625	Jan 08n	107:16	107:17	---	3.85		3.000	Feb 08n	97:26	97:27	---	3.80		5.500	Feb 08n	104:18	104:19	---	3.77		3.375	Feb 08n	98:27	98:28	---	3.80		2.625	May 08n	96:15	96:16	---	3.83															
RATE	MATURITY	BID	ASKED	CHG	ASK	YLD																																																																																																																																																																																																																																																																																																																																																																																																																								
1.625	Apr 05n	99:29	99:30	---	2.67																																																																																																																																																																																																																																																																																																																																																																																																																									
6.500	May 05n	100:10	100:11	---	2.41																																																																																																																																																																																																																																																																																																																																																																																																																									
6.750	May 05n	100:10	100:11	---	2.48																																																																																																																																																																																																																																																																																																																																																																																																																									
12.000	May 05	100:25	100:26	---	2.20																																																																																																																																																																																																																																																																																																																																																																																																																									
1.250	May 05n	99:26	99:27	1	2.45																																																																																																																																																																																																																																																																																																																																																																																																																									
1.125	Jun 05n	99:21	99:22	---	2.59																																																																																																																																																																																																																																																																																																																																																																																																																									
1.500	Jul 05n	99:18	99:19	---	2.81																																																																																																																																																																																																																																																																																																																																																																																																																									
6.500	Aug 05n	101:05	101:06	---	2.89																																																																																																																																																																																																																																																																																																																																																																																																																									
10.750	Aug 05	102:21	---	---	2.81																																																																																																																																																																																																																																																																																																																																																																																																																									
2.000	Aug 05n	99:19	99:20	---	2.95																																																																																																																																																																																																																																																																																																																																																																																																																									
1.625	Sep 05n	99:10	99:11	---	3.06																																																																																																																																																																																																																																																																																																																																																																																																																									
1.625	Oct 05n	99:05	99:06	---	3.14																																																																																																																																																																																																																																																																																																																																																																																																																									
5.750	Nov 05n	101:13	101:14	---	3.22																																																																																																																																																																																																																																																																																																																																																																																																																									
5.875	Nov 05n	101:16	101:17	---	3.21																																																																																																																																																																																																																																																																																																																																																																																																																									
1.875	Nov 05n	99:05	99:06	---	3.19																																																																																																																																																																																																																																																																																																																																																																																																																									
1.875	Dec 05n	99:00	99:01	---	3.24																																																																																																																																																																																																																																																																																																																																																																																																																									
1.875	Jan 06n	98:27	98:28	---	3.29																																																																																																																																																																																																																																																																																																																																																																																																																									
5.625	Feb 06n	101:29	101:30	---	3.26																																																																																																																																																																																																																																																																																																																																																																																																																									
9.375	Feb 06	105:01	105:02	---	3.19																																																																																																																																																																																																																																																																																																																																																																																																																									
1.625	Feb 06n	98:15	98:16	---	3.35																																																																																																																																																																																																																																																																																																																																																																																																																									
1.500	Mar 06n	98:06	98:07	---	3.38																																																																																																																																																																																																																																																																																																																																																																																																																									
2.250	Apr 06n	98:25	98:26	---	3.40																																																																																																																																																																																																																																																																																																																																																																																																																									
2.000	May 06n	98:15	98:16	---	3.40																																																																																																																																																																																																																																																																																																																																																																																																																									
4.625	May 06n	101:08	101:09	---	3.41																																																																																																																																																																																																																																																																																																																																																																																																																									
6.875	May 06n	103:20	103:21	---	3.40																																																																																																																																																																																																																																																																																																																																																																																																																									
2.500	May 06n	98:30	98:31	---	3.42																																																																																																																																																																																																																																																																																																																																																																																																																									
2.750	Jun 06n	99:04	99:05	---	3.45																																																																																																																																																																																																																																																																																																																																																																																																																									
7.000	Jul 06n	104:08	104:09	---	3.48																																																																																																																																																																																																																																																																																																																																																																																																																									
2.750	Jul 06n	99:01	99:02	---	3.48																																																																																																																																																																																																																																																																																																																																																																																																																									
2.375	Aug 06n	98:16	98:17	---	3.51																																																																																																																																																																																																																																																																																																																																																																																																																									
2.375	Aug 06n	98:14	98:15	---	3.52																																																																																																																																																																																																																																																																																																																																																																																																																									
2.500	Sep 06n	98:15	98:16	---	3.55																																																																																																																																																																																																																																																																																																																																																																																																																									
6.500	Oct 06n	104:08	104:09	---	3.55																																																																																																																																																																																																																																																																																																																																																																																																																									
2.500	Oct 06n	98:12	98:13	---	3.57																																																																																																																																																																																																																																																																																																																																																																																																																									
2.625	Nov 06n	98:16	98:17	---	3.57																																																																																																																																																																																																																																																																																																																																																																																																																									
3.500	Nov 06n	99:27	99:28	---	3.57																																																																																																																																																																																																																																																																																																																																																																																																																									
2.875	Nov 06n	98:27	98:28	---	3.58																																																																																																																																																																																																																																																																																																																																																																																																																									
3.000	Dec 06n	98:30	98:31	---	3.61																																																																																																																																																																																																																																																																																																																																																																																																																									
3.375	Jan 07n	105:00	105:01	---	3.61																																																																																																																																																																																																																																																																																																																																																																																																																									
3.125	Jan 07n	99:03	99:04	---	3.62																																																																																																																																																																																																																																																																																																																																																																																																																									
2.250	Feb 07n	97:17	97:18	---	3.62																																																																																																																																																																																																																																																																																																																																																																																																																									
6.250	Feb 07n	104:22	104:23	---	3.57																																																																																																																																																																																																																																																																																																																																																																																																																									
3.375	Feb 07n	99:16	99:17	---	3.63																																																																																																																																																																																																																																																																																																																																																																																																																									
3.750	Mar 07n	100:05	100:06	---	3.65																																																																																																																																																																																																																																																																																																																																																																																																																									
6.625	May 07n	105:27	105:28	---	3.66																																																																																																																																																																																																																																																																																																																																																																																																																									
4.375	May 07n	101:12	101:13	---	3.66																																																																																																																																																																																																																																																																																																																																																																																																																									
3.125	May 07n	98:28	98:29	---	3.67																																																																																																																																																																																																																																																																																																																																																																																																																									
2.750	Aug 07n	97:27	97:28	---	3.71																																																																																																																																																																																																																																																																																																																																																																																																																									
3.250	Aug 07n	98:30	98:31	---	3.71																																																																																																																																																																																																																																																																																																																																																																																																																									
6.125	Aug 07n	105:11	105:12	---	3.70																																																																																																																																																																																																																																																																																																																																																																																																																									
3.000	Nov 07n	98:05	98:06	---	3.74																																																																																																																																																																																																																																																																																																																																																																																																																									
3.625	Jan 08n	107:16	107:17	---	3.85																																																																																																																																																																																																																																																																																																																																																																																																																									
3.000	Feb 08n	97:26	97:27	---	3.80																																																																																																																																																																																																																																																																																																																																																																																																																									
5.500	Feb 08n	104:18	104:19	---	3.77																																																																																																																																																																																																																																																																																																																																																																																																																									
3.375	Feb 08n	98:27	98:28	---	3.80																																																																																																																																																																																																																																																																																																																																																																																																																									
2.625	May 08n	96:15	96:16	---	3.83																																																																																																																																																																																																																																																																																																																																																																																																																									
<table border="1"> <thead> <tr> <th>RATE</th><th>MATURITY</th><th>BID</th><th>ASKED</th><th>CHG</th><th>ASK</th><th>YLD</th></tr> </thead> <tbody> <tr> <td>5.625</td><td>May 08n</td><td>105:04</td><td>105:05</td><td>2</td><td>3.83</td><td></td></tr> <tr> <td>3.250</td><td>Aug 08n</td><td>98:01</td><td>98:02</td><td>2</td><td>3.87</td><td></td></tr> <tr> <td>3.125</td><td>Sep 08n</td><td>97:18</td><td>97:19</td><td>3</td><td>3.88</td><td></td></tr> <tr> <td>3.125</td><td>Oct 08n</td><td>97:15</td><td>97:16</td><td>2</td><td>3.89</td><td></td></tr> <tr> <td>3.750</td><td>Nov 08n</td><td>98:08</td><td>98:09</td><td>2</td><td>3.89</td><td></td></tr> <tr> <td>4.750</td><td>Nov 08n</td><td>102:27</td><td>102:28</td><td>2</td><td>3.88</td><td></td></tr> <tr> <td>1.250</td><td>Nov 08n</td><td>98:06</td><td>98:07</td><td>3</td><td>3.90</td><td></td></tr> <tr> <td>3.250</td><td>Jan 09n</td><td>97:22</td><td>97:23</td><td>2</td><td>3.91</td><td></td></tr> <tr> <td>3.875</td><td>Jan 09n</td><td>110:06</td><td>110:07</td><td>-5</td><td>1.09</td><td></td></tr> <tr> <td>3.000</td><td>Feb 09n</td><td>96:22</td><td>96:23</td><td>2</td><td>3.93</td><td></td></tr> <tr> <td>95:09</td><td>Feb 09n</td><td>95:10</td><td>95:11</td><td>2</td><td>3.92</td><td></td></tr> <tr> <td>97:00</td><td>Apr 09n</td><td>97:00</td><td>97:01</td><td>2</td><td>3.94</td><td></td></tr> <tr> <td>99:23</td><td>May 09n</td><td>99:23</td><td>99:24</td><td>3</td><td>3.94</td><td></td></tr> <tr> <td>105:28</td><td>May 09n</td><td>105:28</td><td>105:29</td><td>2</td><td>3.92</td><td></td></tr> <tr> <td>100:04</td><td>Jun 09n</td><td>100:04</td><td>100:05</td><td>1</td><td>3.95</td><td></td></tr> <tr> <td>98:20</td><td>Jul 09n</td><td>98:20</td><td>98:21</td><td>2</td><td>3.97</td><td></td></tr> <tr> <td>98:05</td><td>Aug 09n</td><td>98:05</td><td>98:06</td><td>2</td><td>3.95</td><td></td></tr> <tr> <td>108:00</td><td>Aug 09n</td><td>108:00</td><td>108:01</td><td>2</td><td>3.96</td><td></td></tr> <tr> <td>97:17</td><td>Sep 09n</td><td>97:17</td><td>97:18</td><td>2</td><td>3.98</td><td></td></tr> <tr> <td>97:14</td><td>Oct 09n</td><td>97:14</td><td>97:15</td><td>3</td><td>3.99</td><td></td></tr> <tr> <td>97:29</td><td>Nov 09n</td><td>97:29</td><td>97:30</td><td>2</td><td>3.99</td><td></td></tr> <tr> <td>97:25</td><td>Dec 09n</td><td>97:25</td><td>97:26</td><td>2</td><td>4.01</td><td></td></tr> <tr> <td>98:08</td><td>Jan 10n</td><td>98:08</td><td>98:09</td><td>2</td><td>4.02</td><td></td></tr> <tr> <td>113:28</td><td>Jan 10n</td><td>113:28</td><td>113:29</td><td>-2</td><td>1.23</td><td></td></tr> <tr> <td>97:23</td><td>Feb 10n</td><td>97:23</td><td>97:24</td><td>2</td><td>4.01</td><td></td></tr> <tr> <td>110:24</td><td>Feb 10n</td><td>110:24</td><td>110:25</td><td>2</td><td>4.02</td><td></td></tr> <tr> <td>99:27</td><td>Mar 10n</td><td>99:27</td><td>99:28</td><td>2</td><td>4.02</td><td></td></tr> <tr> <td>97:31</td><td>Apr 10n</td><td>97:31</td><td>97:32</td><td>-1</td><td>1.30</td><td></td></tr> <tr> <td>100:19</td><td>May 10n</td><td>100:19</td><td>100:20</td><td>-2</td><td>2.40</td><td></td></tr> <tr> <td>108:00</td><td>Aug 10n</td><td>108:00</td><td>108:01</td><td>1</td><td>4.06</td><td></td></tr> <tr> <td>105:14</td><td>Nov 10n</td><td>105:14</td><td>105:15</td><td>-2</td><td>3.21</td><td></td></tr> <tr> <td>111:28</td><td>Jan 11n</td><td>111:28</td><td>111:29</td><td>-2</td><td>1.34</td><td></td></tr> <tr> <td>104:19</td><td>Feb 11n</td><td>104:19</td><td>104:20</td><td>2</td><td>4.10</td><td></td></tr> <tr> <td>111:00</td><td>May 11n</td><td>111:00</td><td>111:01</td><td>-6</td><td>3.43</td><td></td></tr> <tr> <td>104:22</td><td>Aug 11n</td><td>104:22</td><td>104:23</td><td>2</td><td>4.14</td><td></td></tr> <tr> <td>115:30</td><td>Nov 11n</td><td>115:30</td><td>115:31</td><td>1</td><td>3.55</td><td></td></tr> <tr> <td>112:15</td><td>Jan 12n</td><td>112:15</td><td>112:16</td><td>-2</td><td>1.42</td><td></td></tr> <tr> <td>104:02</td><td>Feb 12n</td><td>104:02</td><td>104:03</td><td>1</td><td>4.18</td><td></td></tr> <tr> <td>110:14</td><td>Jul 12n</td><td>110:14</td><td>110:15</td><td>-2</td><td>1.47</td><td></td></tr> <tr> <td>101:03</td><td>Aug 12n</td><td>101:03</td><td>101:04</td><td>---</td><td>4.20</td><td></td></tr> <tr> <td>98:16</td><td>Nov 12n</td><td>98:16</td><td>98:17</td><td>---</td><td>4.23</td><td></td></tr> <tr> <td>116:04</td><td>Nov 12n</td><td>116:04</td><td>116:05</td><td>---</td><td>3.76</td><td></td></tr> <tr> <td>97:12</td><td>Feb 13n</td><td>97:12</td><td>97:13</td><td>-1</td><td>4.26</td><td></td></tr> <tr> <td>95:28</td><td>May 13n</td><td>95:28</td><td>95:29</td><td>-1</td><td>4.23</td><td></td></tr> <tr> <td>102:05</td><td>Jul 13n</td><td>102:05</td><td>102:06</td><td>-1</td><td>1.59</td><td></td></tr> <tr> <td>99:16</td><td>Aug 13n</td><td>99:16</td><td>99:17</td><td>-1</td><td>4.31</td><td></td></tr> <tr> <td>125:06</td><td>Aug 13n</td><td>125:06</td><td>125:07</td><td>-2</td><td>3.87</td><td></td></tr> <tr> <td>99:13</td><td>Nov 13n</td><td>99:13</td><td>99:14</td><td>---</td><td>4.33</td><td></td></tr> <tr> <td>102:25</td><td>Jan 14n</td><td>102:25</td><td>102:26</td><td>-2</td><td>1.65</td><td></td></tr> <tr> <td>97:14</td><td>Feb 14n</td><td>97:14</td><td>97:15</td><td>-1</td><td>4.35</td><td></td></tr> <tr> <td>102:26</td><td>May 14n</td><td>102:26</td><td>102:27</td><td>-2</td><td>4.36</td><td></td></tr> <tr> <td>134:23</td><td>May 14n</td><td>134:23</td><td>134:24</td><td>---</td><td>1.68</td><td></td></tr> <tr> <td>102:22</td><td>Jul 14n</td><td>102:22</td><td>102:23</td><td>---</td><td>1.68</td><td></td></tr> <tr> <td>99:00</td><td>Aug 14n</td><td>99:00</td><td>99:01</td><td>-2</td><td>4.37</td><td></td></tr> <tr> <td>133:18</td><td>Aug 14n</td><td>133:18</td><td>133:19</td><td>---</td><td>3.99</td><td></td></tr> <tr> <td>132:07</td><td>Nov 14n</td><td>132:07</td><td>132:08</td><td>1</td><td>3.98</td><td></td></tr> <tr> <td>98:29</td><td>Nov 14n</td><td>98:29</td><td>98:30</td><td>-1</td><td>4.39</td><td></td></tr> <tr> <td>99:01</td><td>Jan 15n</td><td>99:01</td><td>99:02</td><td>-6</td><td>1.73</td><td></td></tr> </tbody> </table>		RATE	MATURITY	BID	ASKED	CHG	ASK	YLD	5.625	May 08n	105:04	105:05	2	3.83		3.250	Aug 08n	98:01	98:02	2	3.87		3.125	Sep 08n	97:18	97:19	3	3.88		3.125	Oct 08n	97:15	97:16	2	3.89		3.750	Nov 08n	98:08	98:09	2	3.89		4.750	Nov 08n	102:27	102:28	2	3.88		1.250	Nov 08n	98:06	98:07	3	3.90		3.250	Jan 09n	97:22	97:23	2	3.91		3.875	Jan 09n	110:06	110:07	-5	1.09		3.000	Feb 09n	96:22	96:23	2	3.93		95:09	Feb 09n	95:10	95:11	2	3.92		97:00	Apr 09n	97:00	97:01	2	3.94		99:23	May 09n	99:23	99:24	3	3.94		105:28	May 09n	105:28	105:29	2	3.92		100:04	Jun 09n	100:04	100:05	1	3.95		98:20	Jul 09n	98:20	98:21	2	3.97		98:05	Aug 09n	98:05	98:06	2	3.95		108:00	Aug 09n	108:00	108:01	2	3.96		97:17	Sep 09n	97:17	97:18	2	3.98		97:14	Oct 09n	97:14	97:15	3	3.99		97:29	Nov 09n	97:29	97:30	2	3.99		97:25	Dec 09n	97:25	97:26	2	4.01		98:08	Jan 10n	98:08	98:09	2	4.02		113:28	Jan 10n	113:28	113:29	-2	1.23		97:23	Feb 10n	97:23	97:24	2	4.01		110:24	Feb 10n	110:24	110:25	2	4.02		99:27	Mar 10n	99:27	99:28	2	4.02		97:31	Apr 10n	97:31	97:32	-1	1.30		100:19	May 10n	100:19	100:20	-2	2.40		108:00	Aug 10n	108:00	108:01	1	4.06		105:14	Nov 10n	105:14	105:15	-2	3.21		111:28	Jan 11n	111:28	111:29	-2	1.34		104:19	Feb 11n	104:19	104:20	2	4.10		111:00	May 11n	111:00	111:01	-6	3.43		104:22	Aug 11n	104:22	104:23	2	4.14		115:30	Nov 11n	115:30	115:31	1	3.55		112:15	Jan 12n	112:15	112:16	-2	1.42		104:02	Feb 12n	104:02	104:03	1	4.18		110:14	Jul 12n	110:14	110:15	-2	1.47		101:03	Aug 12n	101:03	101:04	---	4.20		98:16	Nov 12n	98:16	98:17	---	4.23		116:04	Nov 12n	116:04	116:05	---	3.76		97:12	Feb 13n	97:12	97:13	-1	4.26		95:28	May 13n	95:28	95:29	-1	4.23		102:05	Jul 13n	102:05	102:06	-1	1.59		99:16	Aug 13n	99:16	99:17	-1	4.31		125:06	Aug 13n	125:06	125:07	-2	3.87		99:13	Nov 13n	99:13	99:14	---	4.33		102:25	Jan 14n	102:25	102:26	-2	1.65		97:14	Feb 14n	97:14	97:15	-1	4.35		102:26	May 14n	102:26	102:27	-2	4.36		134:23	May 14n	134:23	134:24	---	1.68		102:22	Jul 14n	102:22	102:23	---	1.68		99:00	Aug 14n	99:00	99:01	-2	4.37		133:18	Aug 14n	133:18	133:19	---	3.99		132:07	Nov 14n	132:07	132:08	1	3.98		98:29	Nov 14n	98:29	98:30	-1	4.39		99:01	Jan 15n	99:01	99:02	-6	1.73	
RATE	MATURITY	BID	ASKED	CHG	ASK	YLD																																																																																																																																																																																																																																																																																																																																																																																																																								
5.625	May 08n	105:04	105:05	2	3.83																																																																																																																																																																																																																																																																																																																																																																																																																									
3.250	Aug 08n	98:01	98:02	2	3.87																																																																																																																																																																																																																																																																																																																																																																																																																									
3.125	Sep 08n	97:18	97:19	3	3.88																																																																																																																																																																																																																																																																																																																																																																																																																									
3.125	Oct 08n	97:15	97:16	2	3.89																																																																																																																																																																																																																																																																																																																																																																																																																									
3.750	Nov 08n	98:08	98:09	2	3.89																																																																																																																																																																																																																																																																																																																																																																																																																									
4.750	Nov 08n	102:27	102:28	2	3.88																																																																																																																																																																																																																																																																																																																																																																																																																									
1.250	Nov 08n	98:06	98:07	3	3.90																																																																																																																																																																																																																																																																																																																																																																																																																									
3.250	Jan 09n	97:22	97:23	2	3.91																																																																																																																																																																																																																																																																																																																																																																																																																									
3.875	Jan 09n	110:06	110:07	-5	1.09																																																																																																																																																																																																																																																																																																																																																																																																																									
3.000	Feb 09n	96:22	96:23	2	3.93																																																																																																																																																																																																																																																																																																																																																																																																																									
95:09	Feb 09n	95:10	95:11	2	3.92																																																																																																																																																																																																																																																																																																																																																																																																																									
97:00	Apr 09n	97:00	97:01	2	3.94																																																																																																																																																																																																																																																																																																																																																																																																																									
99:23	May 09n	99:23	99:24	3	3.94																																																																																																																																																																																																																																																																																																																																																																																																																									
105:28	May 09n	105:28	105:29	2	3.92																																																																																																																																																																																																																																																																																																																																																																																																																									
100:04	Jun 09n	100:04	100:05	1	3.95																																																																																																																																																																																																																																																																																																																																																																																																																									
98:20	Jul 09n	98:20	98:21	2	3.97																																																																																																																																																																																																																																																																																																																																																																																																																									
98:05	Aug 09n	98:05	98:06	2	3.95																																																																																																																																																																																																																																																																																																																																																																																																																									
108:00	Aug 09n	108:00	108:01	2	3.96																																																																																																																																																																																																																																																																																																																																																																																																																									
97:17	Sep 09n	97:17	97:18	2	3.98																																																																																																																																																																																																																																																																																																																																																																																																																									
97:14	Oct 09n	97:14	97:15	3	3.99																																																																																																																																																																																																																																																																																																																																																																																																																									
97:29	Nov 09n	97:29	97:30	2	3.99																																																																																																																																																																																																																																																																																																																																																																																																																									
97:25	Dec 09n	97:25	97:26	2	4.01																																																																																																																																																																																																																																																																																																																																																																																																																									
98:08	Jan 10n	98:08	98:09	2	4.02																																																																																																																																																																																																																																																																																																																																																																																																																									
113:28	Jan 10n	113:28	113:29	-2	1.23																																																																																																																																																																																																																																																																																																																																																																																																																									
97:23	Feb 10n	97:23	97:24	2	4.01																																																																																																																																																																																																																																																																																																																																																																																																																									
110:24	Feb 10n	110:24	110:25	2	4.02																																																																																																																																																																																																																																																																																																																																																																																																																									
99:27	Mar 10n	99:27	99:28	2	4.02																																																																																																																																																																																																																																																																																																																																																																																																																									
97:31	Apr 10n	97:31	97:32	-1	1.30																																																																																																																																																																																																																																																																																																																																																																																																																									
100:19	May 10n	100:19	100:20	-2	2.40																																																																																																																																																																																																																																																																																																																																																																																																																									
108:00	Aug 10n	108:00	108:01	1	4.06																																																																																																																																																																																																																																																																																																																																																																																																																									
105:14	Nov 10n	105:14	105:15	-2	3.21																																																																																																																																																																																																																																																																																																																																																																																																																									
111:28	Jan 11n	111:28	111:29	-2	1.34																																																																																																																																																																																																																																																																																																																																																																																																																									
104:19	Feb 11n	104:19	104:20	2	4.10																																																																																																																																																																																																																																																																																																																																																																																																																									
111:00	May 11n	111:00	111:01	-6	3.43																																																																																																																																																																																																																																																																																																																																																																																																																									
104:22	Aug 11n	104:22	104:23	2	4.14																																																																																																																																																																																																																																																																																																																																																																																																																									
115:30	Nov 11n	115:30	115:31	1	3.55																																																																																																																																																																																																																																																																																																																																																																																																																									
112:15	Jan 12n	112:15	112:16	-2	1.42																																																																																																																																																																																																																																																																																																																																																																																																																									
104:02	Feb 12n	104:02	104:03	1	4.18																																																																																																																																																																																																																																																																																																																																																																																																																									
110:14	Jul 12n	110:14	110:15	-2	1.47																																																																																																																																																																																																																																																																																																																																																																																																																									
101:03	Aug 12n	101:03	101:04	---	4.20																																																																																																																																																																																																																																																																																																																																																																																																																									
98:16	Nov 12n	98:16	98:17	---	4.23																																																																																																																																																																																																																																																																																																																																																																																																																									
116:04	Nov 12n	116:04	116:05	---	3.76																																																																																																																																																																																																																																																																																																																																																																																																																									
97:12	Feb 13n	97:12	97:13	-1	4.26																																																																																																																																																																																																																																																																																																																																																																																																																									
95:28	May 13n	95:28	95:29	-1	4.23																																																																																																																																																																																																																																																																																																																																																																																																																									
102:05	Jul 13n	102:05	102:06	-1	1.59																																																																																																																																																																																																																																																																																																																																																																																																																									
99:16	Aug 13n	99:16	99:17	-1	4.31																																																																																																																																																																																																																																																																																																																																																																																																																									
125:06	Aug 13n	125:06	125:07	-2	3.87																																																																																																																																																																																																																																																																																																																																																																																																																									
99:13	Nov 13n	99:13	99:14	---	4.33																																																																																																																																																																																																																																																																																																																																																																																																																									
102:25	Jan 14n	102:25	102:26	-2	1.65																																																																																																																																																																																																																																																																																																																																																																																																																									
97:14	Feb 14n	97:14	97:15	-1	4.35																																																																																																																																																																																																																																																																																																																																																																																																																									
102:26	May 14n	102:26	102:27	-2	4.36																																																																																																																																																																																																																																																																																																																																																																																																																									
134:23	May 14n	134:23	134:24	---	1.68																																																																																																																																																																																																																																																																																																																																																																																																																									
102:22	Jul 14n	102:22	102:23	---	1.68																																																																																																																																																																																																																																																																																																																																																																																																																									
99:00	Aug 14n	99:00	99:01	-2	4.37																																																																																																																																																																																																																																																																																																																																																																																																																									
133:18	Aug 14n	133:18	133:19	---	3.99																																																																																																																																																																																																																																																																																																																																																																																																																									
132:07	Nov 14n	132:07	132:08	1	3.98																																																																																																																																																																																																																																																																																																																																																																																																																									
98:29	Nov 14n	98:29	98:30	-1	4.39																																																																																																																																																																																																																																																																																																																																																																																																																									
99:01	Jan 15n	99:01	99:02	-6	1.73																																																																																																																																																																																																																																																																																																																																																																																																																									
<table border="1"> <thead> <tr> <th>RATE</th><th>MATURITY</th><th>BID</th><th>ASKED</th><th>CHG</th><th>ASK</th><th>YLD</th></tr> </thead> <tbody> <tr> <td>4.000</td><td>Feb 15n</td><td>97:00</td><td>97:01</td><td>-2</td><td>4.39</td><td></td></tr> <tr> <td>11:250</td><td>Feb 15</td><td>154:10</td><td>154:11</td><td>-2</td><td>4.39</td><td></td></tr> <tr> <td>10:825</td><td>Aug 15</td><td>151:01</td><td>151:02</td><td>-3</td><td>4.42</td><td></td></tr> <tr> <td>9:875</td><td>Nov 15</td><td>145:13</td><td>145:14</td><td>-3</td><td>4.44</td><td></td></tr> <tr> <td>9:250</td><td>Feb 16</td><td>140:24</td><td>140:25</td><td>-2</td><td>4.46</td><td></td></tr> <tr> <td>7:250</td><td>May 16</td><td>123:23</td><td>123:24</td><td>-4</td><td>4.50</td><td></td></tr> <tr> <td>7:500</td><td>Nov 16</td><td>126:17</td><td>126:18</td><td>-4</td><td>4.53</td><td></td></tr> <tr> <td>8:750</td><td>May 17</td><td>138:21</td><td>138:22</td><td>-3</td><td>4.55</td><td></td></tr> <tr> <td>8:875</td><td>Aug 17</td><td>140:10</td><td>140:11</td><td>-3</td><td>4.56</td><td></td></tr> <tr> <td>9:125</td><td>May 18</td><td>144:06</td><td>144:07</td><td>-5</td><td>4.59</td><td></td></tr> <tr> <td>9:000</td><td>Nov 18</td><td>143:24</td><td>143:25</td><td>-4</td><td>4.62</td><td></td></tr> <tr> <td>8:875</td><td>Feb 19</td><td>142:27</td><td>142:28</td><td>-4</td><td>4.64</td><td></td></tr> <tr> <td>8:125</td><td>Aug 19</td><td>135:26</td><td>135:27</td><td>-6</td><td>4.67</td><td></td></tr> <tr> <td>8:500</td><td>Feb 20</td><td>140:14</td><td>140:15</td><td>-6</td><td>4.68</td><td></td></tr> <tr> <td>8:750</td><td>May 20</td><td>143:17</td><td>143:18</td><td>-8</td><td>4.69</td><td></td></tr> <tr> <td>8:750</td><td>Aug 20</td><td>144:00</td><td>144:01</td><td>-8</td><td>4.69</td><td></td></tr> <tr> <td>7:875</td><td>Feb 21</td><td>134:28</td><td>134:29</td><td>-6</td><td>4.72</td><td></td></tr> <tr> <td>8:125</td><td>May 21</td><td>137:30</td><td>137:31</td><td>-7</td><td>4.73</td><td></td></tr> <tr> <td>8:000</td><td>Nov 21</td><td>138:09</td><td>138:10</td><td>-6</td><td>4.73</td><td></td></tr> <tr> <td>8:000</td><td>Nov 21</td><td>137:05</td><td>137:06</td><td>-4</td><td>4.74</td><td></td></tr> <tr> <td>7:250</td><td>Aug 22</td><td>129:05</td><td>129:06</td><td>-5</td><td>4.76</td><td></td></tr> <tr> <td>7:425</td><td>Nov 22</td><td>133:26</td><td>133:27</td><td>-5</td><td>4.76</td><td></td></tr> <tr> <td>7:125</td><td>Feb 23</td><td>128:00</td><td>128:01</td><td>-7</td><td>4.77</td><td></td></tr> <tr> <td>6:250</td><td>Aug 23</td><td>117:23</td><td>117:24</td><td>-5</td><td>4.78</td><td></td></tr> <tr> <td>7:500</td><td>Nov 24</td><td>134:09</td><td>134:10</td><td>-7</td><td>4.78</td><td></td></tr> <tr> <td>2:375</td><td>Jan 25n</td><td>107:25</td><td>107:26</td><td>-8</td><td>1.90</td><td></td></tr> <tr> <td>7:425</td><td>Feb 25</td><td>136:03</td><td>136:04</td><td>-8</td><td>4.78</td><td></td></tr> <tr> <td>6:875</td><td>Aug 25</td><td>126:26</td><td>126:27</td><td>-8</td><td>4.79</td><td></td></tr> <tr> <td>6:000</td><td>Feb 26</td><td>115:20</td><td>115:21</td><td>-8</td><td>4.80</td><td></td></tr> <tr> <td>6:750</td><td>Aug 26</td><td>125:27</td><td>125:28</td><td>-8</td><td>4.80</td><td></td></tr> <tr> <td>6:500</td><td>Nov 26</td><td>122:21</td><td>122:22</td><td>-9</td><td>4.80</td><td></td></tr> <tr> <td>6:625</td><td>Feb 27</td><td>124:16</td><td>124:17</td><td>-9</td><td>4.80</td><td></td></tr> <tr> <td>6:375</td><td>Aug 27</td><td>121:13</td><td>121:14</td><td>-10</td><td>4.80</td><td></td></tr> <tr> <td>6:125</td><td>Nov 27</td><td>118:04</td><td>118:05</td><td>-10</td><td>4.80</td><td></td></tr> <tr> <td>3:625</td><td>Apr 28n</td><td>131:25</td><td>131:26</td><td>-10</td><td>1.91</td><td></td></tr> <tr> <td>5:500</td><td>Aug 28</td><td>109:26</td><td>109:27</td><td>-9</td><td>4.79</td><td></td></tr> <tr> <td>5:250</td><td>Nov 28</td><td>106:11</td><td>106:12</td><td>-9</td><td>4.80</td><td></td></tr> <tr> <td>5:250</td><td>Feb 29</td><td>106:14</td><td>106:15</td><td>-10</td><td>4.79</td><td></td></tr> <tr> <td>3:875</td><td>Apr 29n</td><td>137:27</td><td>137:28</td><td>-12</td><td>1.90</td><td></td></tr> <tr> <td>6:125</td><td>Aug 29</td><td>119:02</td><td>119:03</td><td>-10</td><td>4.79</td><td></td></tr> <tr> <td>6:250</td><td>May 30</td><td>121:14</td><td>121:15</td><td>-12</td><td>4.77</td><td></td></tr> <tr> <td>5:375</td><td>Feb 31</td><td>110:08</td><td>110:09</td><td>-11</td><td>4.68</td><td></td></tr> <tr> <td>5:375</td><td>Apr 32n</td><td>133:02</td><td>133:03</td><td>-8</td><td>1.82</td><td></td></tr> </tbody> </table>		RATE	MATURITY	BID	ASKED	CHG	ASK	YLD	4.000	Feb 15n	97:00	97:01	-2	4.39		11:250	Feb 15	154:10	154:11	-2	4.39		10:825	Aug 15	151:01	151:02	-3	4.42		9:875	Nov 15	145:13	145:14	-3	4.44		9:250	Feb 16	140:24	140:25	-2	4.46		7:250	May 16	123:23	123:24	-4	4.50		7:500	Nov 16	126:17	126:18	-4	4.53		8:750	May 17	138:21	138:22	-3	4.55		8:875	Aug 17	140:10	140:11	-3	4.56		9:125	May 18	144:06	144:07	-5	4.59		9:000	Nov 18	143:24	143:25	-4	4.62		8:875	Feb 19	142:27	142:28	-4	4.64		8:125	Aug 19	135:26	135:27	-6	4.67		8:500	Feb 20	140:14	140:15	-6	4.68		8:750	May 20	143:17	143:18	-8	4.69		8:750	Aug 20	144:00	144:01	-8	4.69		7:875	Feb 21	134:28	134:29	-6	4.72		8:125	May 21	137:30	137:31	-7	4.73		8:000	Nov 21	138:09	138:10	-6	4.73		8:000	Nov 21	137:05	137:06	-4	4.74		7:250	Aug 22	129:05	129:06	-5	4.76		7:425	Nov 22	133:26	133:27	-5	4.76		7:125	Feb 23	128:00	128:01	-7	4.77		6:250	Aug 23	117:23	117:24	-5	4.78		7:500	Nov 24	134:09	134:10	-7	4.78		2:375	Jan 25n	107:25	107:26	-8	1.90		7:425	Feb 25	136:03	136:04	-8	4.78		6:875	Aug 25	126:26	126:27	-8	4.79		6:000	Feb 26	115:20	115:21	-8	4.80		6:750	Aug 26	125:27	125:28	-8	4.80		6:500	Nov 26	122:21	122:22	-9	4.80		6:625	Feb 27	124:16	124:17	-9	4.80		6:375	Aug 27	121:13	121:14	-10	4.80		6:125	Nov 27	118:04	118:05	-10	4.80		3:625	Apr 28n	131:25	131:26	-10	1.91		5:500	Aug 28	109:26	109:27	-9	4.79		5:250	Nov 28	106:11	106:12	-9	4.80		5:250	Feb 29	106:14	106:15	-10	4.79		3:875	Apr 29n	137:27	137:28	-12	1.90		6:125	Aug 29	119:02	119:03	-10	4.79		6:250	May 30	121:14	121:15	-12	4.77		5:375	Feb 31	110:08	110:09	-11	4.68		5:375	Apr 32n	133:02	133:03	-8	1.82																																																																																																										
RATE	MATURITY	BID	ASKED	CHG	ASK	YLD																																																																																																																																																																																																																																																																																																																																																																																																																								
4.000	Feb 15n	97:00	97:01	-2	4.39																																																																																																																																																																																																																																																																																																																																																																																																																									
11:250	Feb 15	154:10	154:11	-2	4.39																																																																																																																																																																																																																																																																																																																																																																																																																									
10:825	Aug 15	151:01	151:02	-3	4.42																																																																																																																																																																																																																																																																																																																																																																																																																									
9:875	Nov 15	145:13	145:14	-3	4.44																																																																																																																																																																																																																																																																																																																																																																																																																									
9:250	Feb 16	140:24	140:25	-2	4.46																																																																																																																																																																																																																																																																																																																																																																																																																									
7:250	May 16	123:23	123:24	-4	4.50																																																																																																																																																																																																																																																																																																																																																																																																																									
7:500	Nov 16	126:17	126:18	-4	4.53																																																																																																																																																																																																																																																																																																																																																																																																																									
8:750	May 17	138:21	138:22	-3	4.55																																																																																																																																																																																																																																																																																																																																																																																																																									
8:875	Aug 17	140:10	140:11	-3	4.56																																																																																																																																																																																																																																																																																																																																																																																																																									
9:125	May 18	144:06	144:07	-5	4.59																																																																																																																																																																																																																																																																																																																																																																																																																									
9:000	Nov 18	143:24	143:25	-4	4.62																																																																																																																																																																																																																																																																																																																																																																																																																									
8:875	Feb 19	142:27	142:28	-4	4.64																																																																																																																																																																																																																																																																																																																																																																																																																									
8:125	Aug 19	135:26	135:27	-6	4.67																																																																																																																																																																																																																																																																																																																																																																																																																									
8:500	Feb 20	140:14	140:15	-6	4.68																																																																																																																																																																																																																																																																																																																																																																																																																									
8:750	May 20	143:17	143:18	-8	4.69																																																																																																																																																																																																																																																																																																																																																																																																																									
8:750	Aug 20	144:00	144:01	-8	4.69																																																																																																																																																																																																																																																																																																																																																																																																																									
7:875	Feb 21	134:28	134:29	-6	4.72																																																																																																																																																																																																																																																																																																																																																																																																																									
8:125	May 21	137:30	137:31	-7	4.73																																																																																																																																																																																																																																																																																																																																																																																																																									
8:000	Nov 21	138:09	138:10	-6	4.73																																																																																																																																																																																																																																																																																																																																																																																																																									
8:000	Nov 21	137:05	137:06	-4	4.74																																																																																																																																																																																																																																																																																																																																																																																																																									
7:250	Aug 22	129:05	129:06	-5	4.76																																																																																																																																																																																																																																																																																																																																																																																																																									
7:425	Nov 22	133:26	133:27	-5	4.76																																																																																																																																																																																																																																																																																																																																																																																																																									
7:125	Feb 23	128:00	128:01	-7	4.77																																																																																																																																																																																																																																																																																																																																																																																																																									
6:250	Aug 23	117:23	117:24	-5	4.78																																																																																																																																																																																																																																																																																																																																																																																																																									
7:500	Nov 24	134:09	134:10	-7	4.78																																																																																																																																																																																																																																																																																																																																																																																																																									
2:375	Jan 25n	107:25	107:26	-8	1.90																																																																																																																																																																																																																																																																																																																																																																																																																									
7:425	Feb 25	136:03	136:04	-8	4.78																																																																																																																																																																																																																																																																																																																																																																																																																									
6:875	Aug 25	126:26	126:27	-8	4.79																																																																																																																																																																																																																																																																																																																																																																																																																									
6:000	Feb 26	115:20	115:21	-8	4.80																																																																																																																																																																																																																																																																																																																																																																																																																									
6:750	Aug 26	125:27	125:28	-8	4.80																																																																																																																																																																																																																																																																																																																																																																																																																									
6:500	Nov 26	122:21	122:22	-9	4.80																																																																																																																																																																																																																																																																																																																																																																																																																									
6:625	Feb 27	124:16	124:17	-9	4.80																																																																																																																																																																																																																																																																																																																																																																																																																									
6:375	Aug 27	121:13	121:14	-10	4.80																																																																																																																																																																																																																																																																																																																																																																																																																									
6:125	Nov 27	118:04	118:05	-10	4.80																																																																																																																																																																																																																																																																																																																																																																																																																									
3:625	Apr 28n	131:25	131:26	-10	1.91																																																																																																																																																																																																																																																																																																																																																																																																																									
5:500	Aug 28	109:26	109:27	-9	4.79																																																																																																																																																																																																																																																																																																																																																																																																																									
5:250	Nov 28	106:11	106:12	-9	4.80																																																																																																																																																																																																																																																																																																																																																																																																																									
5:250	Feb 29	106:14	106:15	-10	4.79																																																																																																																																																																																																																																																																																																																																																																																																																									
3:875	Apr 29n	137:27	137:28	-12	1.90																																																																																																																																																																																																																																																																																																																																																																																																																									
6:125	Aug 29	119:02	119:03	-10	4.79																																																																																																																																																																																																																																																																																																																																																																																																																									
6:250	May 30	121:14	121:15	-12	4.77																																																																																																																																																																																																																																																																																																																																																																																																																									
5:375	Feb 31	110:08	110:09	-11	4.68																																																																																																																																																																																																																																																																																																																																																																																																																									
5:375	Apr 32n	133:02	133:03	-8	1.82																																																																																																																																																																																																																																																																																																																																																																																																																									
<table border="1"> <thead> <tr> <th>RATE</th><th>MATURITY</th><th>BID</th><th>ASKED</th><th>CHG</th><th>ASK</th><th>YLD</th></tr> </thead> <tbody> <tr> <td>2.50</td><td>Apr 21 05</td><td>7</td><td>2.59</td><td>2.58</td><td>-0.02</td><td>2.62</td></tr> <tr> <td>2.55</td><td>Apr 28 05</td><td>21</td><td>2.55</td><td>2.54</td><td>0.01</td><td>2.58</td></tr> <tr> <td>2.57</td><td>May 05 05</td><td>21</td><td>2.57</td><td>2.56</td><td>0.01</td><td>2.60</td></tr> <tr> <td>2.58</td><td>May 12 05</td><td>28</td><td>2.58</td><td>2.57</td><td>0.02</td><td>2.61</td></tr> <tr> <td>2.55</td><td>May 19 05</td><td>35</td><td>2.55</td><td>2.54</td><td>-0.03</td><td>2.58</td></tr> <tr> <td>2.59</td><td>May 26 05</td><td>42</td><td>2.59</td><td>2.58</td><td>0.01</td><td>2.62</td></tr> <tr> <td>2.60</td><td>Jun 02 05</td><td>49</td><td>2.61</td><td>2.60</td><td>0.01</td><td>2.65</td></tr> <tr> <td>2.63</td><td>Jun 09 05</td><td>56</td><td>2.64</td><td>2.63</td><td>0.01</td><td>2.68</td></tr> <tr> <td>2.64</td><td>Jun 16 05</td><td>63</td><td>2.65</td><td>2.64</td><td>0.01</td><td>2.69</td></tr> <tr> <td>2.68</td><td>Jun 23 05</td><td>70</td><td>2.68</td><td>2.67</td><td>0.02</td><td>2.72</td></tr> <tr> <td>2.69</td><td>Jun 30 05</td><td>77</td><td>2.69</td><td>2.68</td><td>0.02</td><td>2.73</td></tr> <tr> <td>2.70</td><td>Jul 07 05</td><td>84</td><td>2.71</td><td>2.70</td><td>---</td><td>2.75</td></tr> <tr> <td>2.71</td><td>Jul 14 05</td><td>91</td><td>2.71</td><td>2.70</td><td>---</td><td>2.76</td></tr> <tr> <td>2.74</td><td>Jul 21 05</td><td>98</td><td>2.75</td><td>2.74</td><td>0.01</td><td>2.80</td></tr> <tr> <td>2.75</td><td>Jul 28 05</td><td>105</td><td>2.75</td><td>2.74</td><td>-0.02</td><td>2.80</td></tr> <tr> <td>2.82</td><td>Aug 04 05</td><td>112</td><td>2.82</td><td>2.81</td><td>---</td><td>2.87</td></tr> <tr> <td>2.86</td><td>Aug 11 05</td><td>119</td><td>2.87</td><td>2.86</td><td>0.01</td><td>2.93</td></tr> <tr> <td>2.88</td><td>Aug 18 05</td><td>126</td><td>2.89</td></tr></tbody></table>	RATE	MATURITY	BID	ASKED	CHG	ASK	YLD	2.50	Apr 21 05	7	2.59	2.58	-0.02	2.62	2.55	Apr 28 05	21	2.55	2.54	0.01	2.58	2.57	May 05 05	21	2.57	2.56	0.01	2.60	2.58	May 12 05	28	2.58	2.57	0.02	2.61	2.55	May 19 05	35	2.55	2.54	-0.03	2.58	2.59	May 26 05	42	2.59	2.58	0.01	2.62	2.60	Jun 02 05	49	2.61	2.60	0.01	2.65	2.63	Jun 09 05	56	2.64	2.63	0.01	2.68	2.64	Jun 16 05	63	2.65	2.64	0.01	2.69	2.68	Jun 23 05	70	2.68	2.67	0.02	2.72	2.69	Jun 30 05	77	2.69	2.68	0.02	2.73	2.70	Jul 07 05	84	2.71	2.70	---	2.75	2.71	Jul 14 05	91	2.71	2.70	---	2.76	2.74	Jul 21 05	98	2.75	2.74	0.01	2.80	2.75	Jul 28 05	105	2.75	2.74	-0.02	2.80	2.82	Aug 04 05	112	2.82	2.81	---	2.87	2.86	Aug 11 05	119	2.87	2.86	0.01	2.93	2.88	Aug 18 05	126	2.89																																																																																																																																																																																																																																																																																												
RATE	MATURITY	BID	ASKED	CHG	ASK	YLD																																																																																																																																																																																																																																																																																																																																																																																																																								
2.50	Apr 21 05	7	2.59	2.58	-0.02	2.62																																																																																																																																																																																																																																																																																																																																																																																																																								
2.55	Apr 28 05	21	2.55	2.54	0.01	2.58																																																																																																																																																																																																																																																																																																																																																																																																																								
2.57	May 05 05	21	2.57	2.56	0.01	2.60																																																																																																																																																																																																																																																																																																																																																																																																																								
2.58	May 12 05	28	2.58	2.57	0.02	2.61																																																																																																																																																																																																																																																																																																																																																																																																																								
2.55	May 19 05	35	2.55	2.54	-0.03	2.58																																																																																																																																																																																																																																																																																																																																																																																																																								
2.59	May 26 05	42	2.59	2.58	0.01	2.62																																																																																																																																																																																																																																																																																																																																																																																																																								
2.60	Jun 02 05	49	2.61	2.60	0.01	2.65																																																																																																																																																																																																																																																																																																																																																																																																																								
2.63	Jun 09 05	56	2.64	2.63	0.01	2.68																																																																																																																																																																																																																																																																																																																																																																																																																								
2.64	Jun 16 05	63	2.65	2.64	0.01	2.69																																																																																																																																																																																																																																																																																																																																																																																																																								
2.68	Jun 23 05	70	2.68	2.67	0.02	2.72																																																																																																																																																																																																																																																																																																																																																																																																																								
2.69	Jun 30 05	77	2.69	2.68	0.02	2.73																																																																																																																																																																																																																																																																																																																																																																																																																								
2.70	Jul 07 05	84	2.71	2.70	---	2.75																																																																																																																																																																																																																																																																																																																																																																																																																								
2.71	Jul 14 05	91	2.71	2.70	---	2.76																																																																																																																																																																																																																																																																																																																																																																																																																								
2.74	Jul 21 05	98	2.75	2.74	0.01	2.80																																																																																																																																																																																																																																																																																																																																																																																																																								
2.75	Jul 28 05	105	2.75	2.74	-0.02	2.80																																																																																																																																																																																																																																																																																																																																																																																																																								
2.82	Aug 04 05	112	2.82	2.81	---	2.87																																																																																																																																																																																																																																																																																																																																																																																																																								
2.86	Aug 11 05	119	2.87	2.86	0.01	2.93																																																																																																																																																																																																																																																																																																																																																																																																																								
2.88	Aug 18 05	126	2.89																																																																																																																																																																																																																																																																																																																																																																																																																											

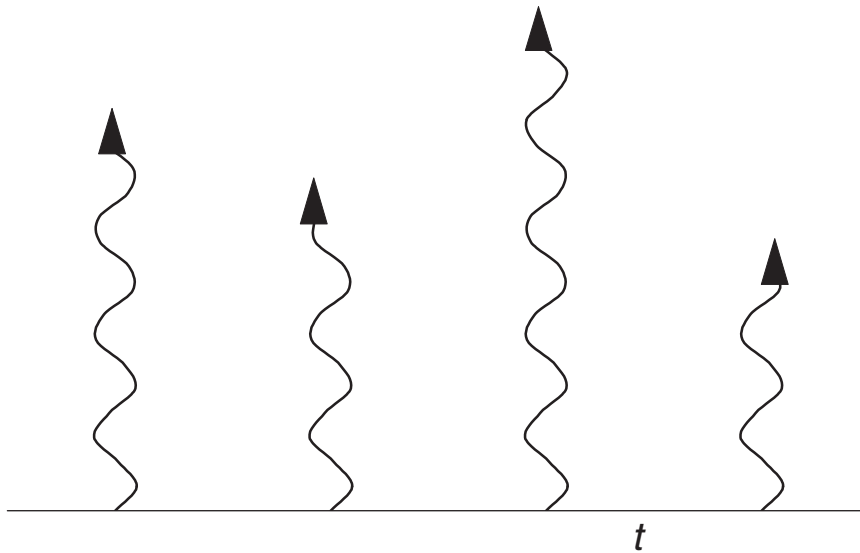


Figure 14.4 The floating-rate bond.

The most common measure of interest is **London Interbank Offer Rate** or **LIBOR**. LIBOR comes in various maturities, one month, three month, six month, etc., and is the rate of interest offered between Eurocurrency banks for fixed-term deposits.

Sometimes the coupon payment on a bond is not a prescribed dollar amount but depends on the level of some ‘index,’ measured at the time of the payment or before. Typically, we cannot know at the start of the contract what level this index will be at when the payment is made. We will see examples of such contracts in later chapters.

14.2.5 Forward rate agreements

A **forward rate agreement (FRA)** is an agreement between two parties that a prescribed interest rate will apply to a prescribed principal over some specified period in the future. The cashflows in this agreement are as follows: party A pays party B the principal at time T_1 and B pays A the principal plus agreed interest at time $T_2 > T_1$. The value of this exchange at the time the contract is entered into is generally not zero and so there will be a transfer of cash from one party to the other at the start date.

14.2.6 Repos

A **repo** is a repurchase agreement. It is an agreement to sell some security to another party and buy it back at a fixed date and for a fixed amount. The price at which the security is bought back is greater than the selling price and the difference implies an interest rate called the **repo rate**. The commonest repo is the overnight repo in which the agreement is renegotiated daily. If the repo agreement extends for 30 days it is called a **term repo**.

A **reverse repo** is the borrowing of a security for a short period at an agreed interest rate.

Repos can be used to lock in future interest rates. For example, buy a six-month Treasury bill today and repo it out for three months. There is no cash flow today since the bond has been paid for (money out) and then repoed (same amount in). In three months' time you will have to repurchase the bill at the agreed price, this is an outflow of cash. In six months you receive the principal. Money out in three months, money in in six months, for there to be no arbitrage the equivalent interest rate should be that currently prevailing between three and six months' time.

14.2.7 STRIPS

STRIPS stands for Separate Trading of Registered Interest and Principal of Securities. The coupons and principal of normal bonds are split up, creating artificial zero-coupon bonds of longer maturity than would otherwise be available.

14.2.8 Amortization

In all of the above products I have assumed that the principal remains fixed at its initial level. Sometimes this is not the case, the principal can **amortize** or decrease during the life of the contract. The principal is thus paid back gradually and interest is paid on the amount of the principal outstanding. Such amortization is arranged at the initiation of the contract and may be fixed, so that the rate of decrease of the principal is known beforehand, or can depend on the level of some index, if the index is high the principal amortizes faster, for example. We see an example of a complex amortizing structure in Chapter 18.

14.2.9 Call provision

Some bonds have a **call provision**. The issuer can call back the bond on certain dates or at certain periods for a prescribed, possibly time-dependent, amount. This lowers the value of the bond. The mathematical consequences of this are discussed in Chapter 16.

14.3 INTERNATIONAL BOND MARKETS

14.3.1 United States of America

In the US, bonds of maturity less than one year are called **bills** and are usually zero coupon. Bonds with maturity 2–10 years are called **notes**. They are coupon bearing with coupons every six months. Bonds with maturity greater than 10 years are called **bonds**. Again they are coupon bearing. In this book I tend to call all of these 'bonds,' merely specifying whether or not they have coupons.

Bonds traded in the United States foreign bond market but which are issued by non-US institutions are called **Yankee** bonds.

Since the beginning of 1997 the US government has also issued bonds linked to the rate of inflation.

14.3.2 United Kingdom

Bonds issued by the UK government are called **gilts**. Some of these bonds are callable, some are irredeemable, meaning that they are perpetual bonds having a coupon but no repayment of principal. The government also issues convertible bonds which may be converted into another bond issue, typically of longer maturity. Finally, there are index-linked bonds having the amount of the coupon and principal payments linked to a measure of inflation, the Retail Price Index (RPI).

14.3.3 Japan

Japanese government bonds (JGBs) come as short-term treasury bills, medium-term, long-term (10-year maturity) and super long-term (20-year maturity). The long- and super long-term bonds have coupons every six months. The short-term bonds have no coupons and the medium-term bonds can be either coupon-bearing or zero-coupon bonds.

Yen denominated bonds issued by non-Japanese institutions are called **Samurai** bonds.

14.4 ACCRUED INTEREST

The market price of bonds quoted in the newspapers are **clean prices**. That is, they are quoted without any **accrued interest**. The accrued interest is the amount of interest that has built up since the last coupon payment:

$$\begin{aligned} \text{accrued interest} &= \text{interest due in full period} \\ &\times \frac{\text{number of days since last coupon date}}{\text{number of days in period between coupon payments}}. \end{aligned}$$

The actual payment is called the **dirty price** and is the sum of the quoted clean price and the accrued interest.

14.5 DAY-COUNT CONVENTIONS

Because of such matters as the accrual of interest between coupon dates there naturally arises the question of how to accrue interest over shorter periods. Interest is accrued between two dates according to the formula

$$\frac{\text{number of days between the two dates}}{\text{number of days in period}} \times \text{interest earned in reference period}.$$

There are three main ways of calculating the 'number of days' in the above expression.

- **Actual/Actual** Simply count the number of calendar days.
- **30/360** Assume there are 30 days in a month and 360 days in a year.
- **Actual/360** Each month has the right number of days but there are only 360 days in a year.

14.6 CONTINUOUSLY AND DISCRETELY COMPOUNDED INTEREST

To be able to compare fixed-income products we must decide on a convention for the measurement of interest rates. So far, we have used a continuously compounded rate, meaning that the present value of \$1 paid at time T in the future is

$$e^{-rT} \times \$1$$

for some r . We have seen how this follows from the cash-in-the-bank or money market account equation

$$dM = rM dt.$$

This is the convention used in the options world.

Another common convention is to use the formula

$$\frac{1}{(1+r')^T} \times \$1,$$

for present value, where r' is some interest rate. This represents discretely compounded interest and assumes that interest is accumulated *annually* for T years. The formula is derived from calculating the present value from a single-period payment, and then compounding this for each year. This formula is commonly used for the simpler type of instruments such as coupon-bearing bonds. The two formulæ are identical, of course, when

$$r = \log(1 + r').$$

This gives the relationship between the continuously compounded interest rate r and the discrete version r' . What would the formula be if interest was discretely compounded twice per year?

In this book we tend to use the continuous definition of interest rates.



Time Out...

Which is better?

Would you rather get 10% once a year or 5% twice a year? With the former \$1 would be worth \$1.10 at the end of the year, whereas the two installments of 5% would give you

$$(1 + 0.05)^2 = 1.1025.$$

It is very important to know what kind of interest payment you are getting.

What interest rate paid continuously is equivalent to a one-off 10%? The answer is the r which satisfies

$$e^r = 1.1$$

i.e.

$$r = \log(1.1) = 0.09531018 \dots$$

So, about 9.53% on an annualized basis. Remember this logarithm is the natural logarithm, also denoted (as in Excel) by $\ln(\cdot)$.

14.7 MEASURES OF YIELD

There is such a variety of fixed-income products, with different coupon structures, amortization, fixed and/or floating rates, that it is necessary to be able to consistently compare different products. Suppose you have to choose between a 10-year zero-coupon bond and a 20-year coupon-bearing bond. One has no income for 10 years but then gets a big lump sum, the other has a trickle of income but you have to wait much longer for the big amount.

One way to do this is through measures of how much each contract earns, there are several measures of this all coming under the name **yield**.

14.7.1 Current yield

The simplest measurement of how much a contract earns is the **current yield**. This measure is defined by

$$\text{current yield} = \frac{\text{annual \$ coupon income}}{\text{bond price}}.$$

For example, consider the 10-year bond that pays 2 cents every six months and \$1 at maturity. This bond has a total income per annum of 4 cents. Suppose that the quoted market price of this bond is 88 cents. The current yield is simply

$$\frac{0.04}{0.88} = 4.5\%.$$

This measurement of the yield of the bond makes no allowance for the payment of the principal at maturity, nor for the time value of money if the coupon payment is reinvested, nor for any capital gain or loss that may be made if the bond is sold before maturity. It is a relatively unsophisticated measure, concentrating very much on short-term properties of the bond.

14.7.2 The yield to maturity (YTM) or internal rate of return (IRR)

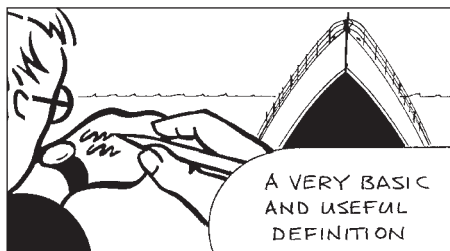
Suppose that we have a zero-coupon bond maturing at time T when it pays one dollar. At time t it has a value $Z(t; T)$. Applying a constant rate of return of y between t and T , then

one dollar received at time T has a present value of $Z(t; T)$ at time t , where

$$Z(t; T) = e^{-y(T-t)}.$$

It follows that

$$y = -\frac{\log Z}{T-t}.$$



Let us generalize this. Suppose that we have a coupon-bearing bond. Discount all coupons and the principal to the present by using some interest rate y . The present value of the bond, at time t , is then

$$V = Pe^{-y(T-t)} + \sum_{i=1}^N C_i e^{-y(t_i-t)}, \quad (14.1)$$

where P is the principal, N the number of coupons, C_i the coupon paid on date t_i .

If the bond is a traded security then we know the price at which the bond can be bought. If this is the case then we can calculate the **yield to maturity** or **internal rate of return** as the value y that we must put into Equation (14.1) to make V equal to the traded price of the bond. This calculation must be performed by some trial and error/iterative procedure. For example, in the bond in Table 14.1 we have a principal of \$1 paid in five years and coupons of three cents (three percent) paid every six months.

Suppose that the market value of this bond is 96 cents. We ask 'What is the internal rate of return we must use to give these cash flows a total present value of 96 cents?' This value is the yield to maturity. In the fourth column in this table is the present value (PV) of each of the cashflows using a rate of 6.8406%: since the sum of these present values is 96 cents the YTM or IRR is 6.8406%.

This yield to maturity is a valid measure of the return on a bond if we intend to hold it to maturity.

Table 14.1 An example of a coupon-bearing bond.

Time	Coupon	Principal repayment	PV (discounting at 6.8406%)
0			0
0.5	.03		0.0290
1.0	.03		0.0280
1.5	.03		0.0270
2.0	.03		0.0262
2.5	.03		0.0253
3.0	.03		0.0244
3.5	.03		0.0236
4.0	.03		0.0228
4.5	.03		0.0220
5.0	.03	1.00	0.7316
Total			0.9600

To calculate the yield to maturity of a portfolio of bonds simply treat all the cashflows as if they were from the one bond and calculate the value of the whole portfolio by adding up the market values of the component bonds.

14.8 THE YIELD CURVE

The plot of yield to maturity against time to maturity is called the **yield curve**. For the moment assume that this has been calculated from zero-coupon bonds and that these bonds have been issued by a perfectly creditworthy source.

If the bonds have coupons then the calculation of the yield curve is more complicated and the 'forward curve,' described below, is a better measure of the interest rate pertaining at some time in the future. Figure 14.5 shows the yield curve for US Treasuries as it was on 9th September 1999.

Time Out...

Discount factors

Once you've calculated the yield curve you can use the results to work out the present value of any fixed-rate cashflows. All you have to do is work out the discount factor for each cashflow and multiply the cashflow by that amount. Typically, you'll find yourself in the situation of having a cashflow on a certain date but no yield associated with that maturity. Then you'll have to interpolate between the two yields either side to get an estimate for the required maturity.



14.9 PRICE/YIELD RELATIONSHIP

From Equation (14.1) we can easily see that the relationship between the price of a bond and its yield is of the form shown in Figure 14.6 (assuming that all cash flows are positive). On this figure is marked the current market price and the current yield to maturity.

Since we are often interested in the sensitivity of instruments to the movement of certain underlying factors it is natural to ask how does the price of a bond vary with the yield, or vice versa. To a first approximation this variation can be quantified by a measure called the duration.

Figure 14.7 shows the price/yield relationship for a specific five-year US Treasury.



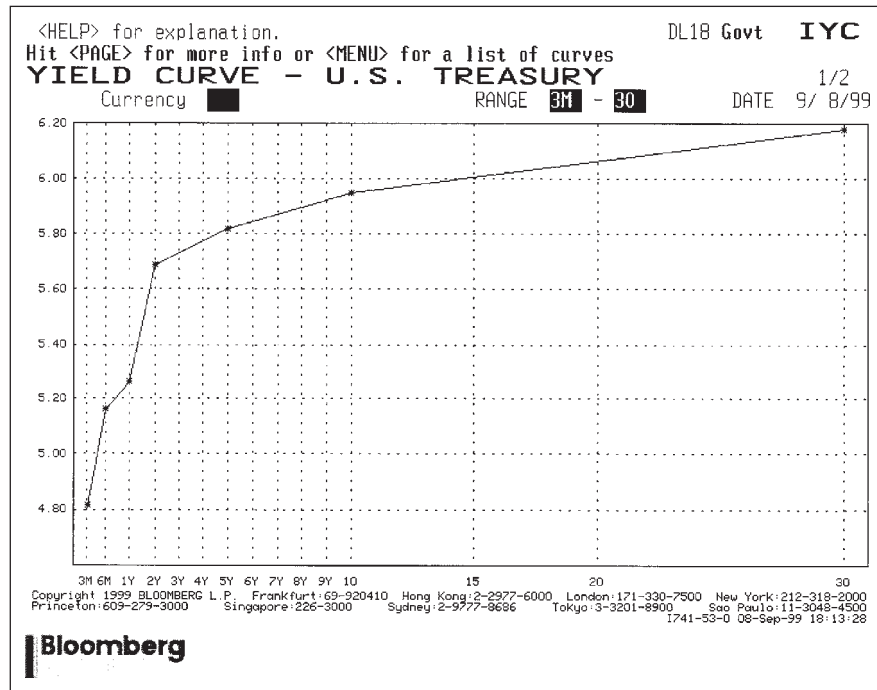


Figure 14.5 Yield curve for US Treasuries. Source: Bloomberg L.P.

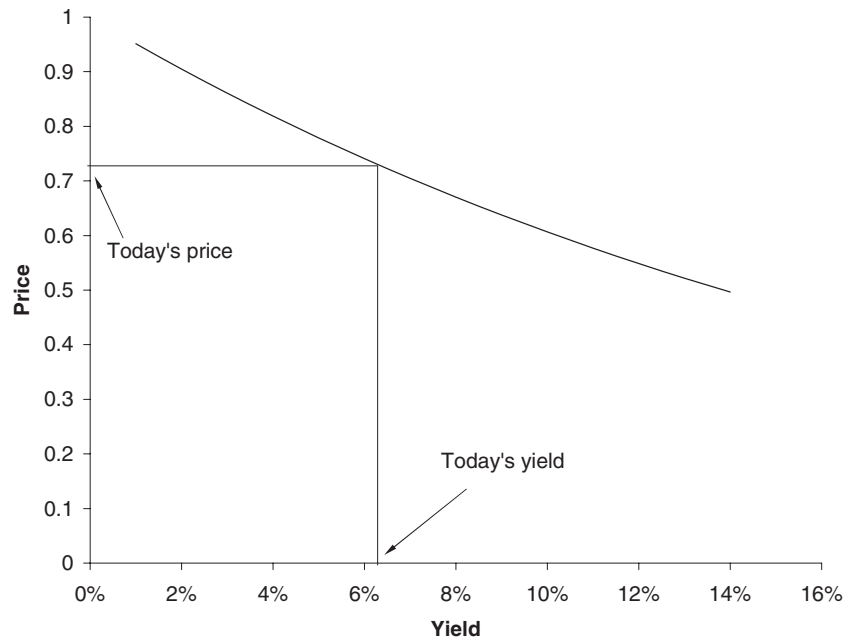


Figure 14.6 The price/yield relationship.

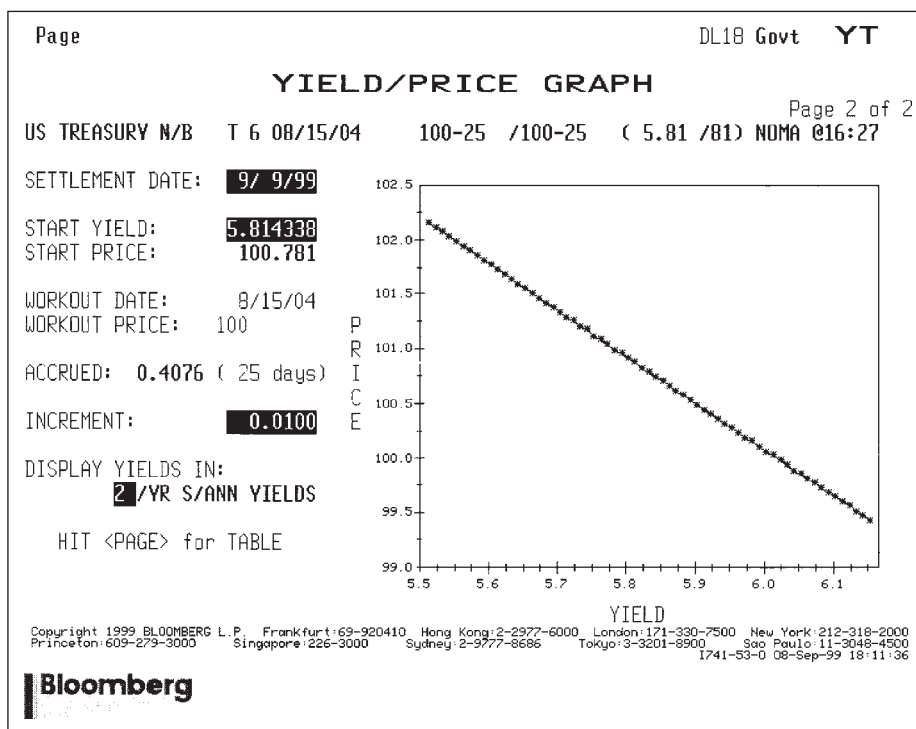


Figure 14.7 Bloomberg's price/yield graph. Source: Bloomberg L.P.

14.10 DURATION

From Equation (14.1) we find that

$$\frac{dV}{dy} = -(T-t)Pe^{-y(T-t)} - \sum_{i=1}^N C_i(t_i-t)e^{-y(t_i-t)}.$$

This is the slope of the price/yield curve. The quantity

$$-\frac{1}{V} \frac{dV}{dy}$$

is called the **Macaulay duration**. (The **modified duration** is similar but uses the discretely compounded rate.) In the expression for the duration the time of each coupon payment is weighted by its present value. The higher the value of the present value of the coupon the more it contributes to the duration. Also, since y is measured in units of inverse time, the units of the duration are time. The duration is a measure of the average life of the bond. It is easily shown that the Macaulay duration for a zero-coupon bond is the same as its maturity.

Let's take a look at the idea of average time. Suppose we asked what zero-coupon bond is our (coupon-bearing) bond equivalent to? That is, what maturity would an 'equivalent' bond have? Take the actual bond's value and equate it to a zero-coupon bond, having





All of this is
implemented on
the CD

the same yield but an unknown maturity (and unknown quantity!):

$$V = Pe^{-y(T-t)} + \sum_{i=1}^N C_i e^{-y(t_i-t)} = X e^{-y(\bar{T}-t)}.$$

Differentiate both sides with respect to y :

$$\frac{dV}{dy} = -(T-t)Pe^{-y(T-t)} - \sum_{i=1}^N C_i(t_i-t)e^{-y(t_i-t)} = -X(\bar{T}-t)e^{-y(\bar{T}-t)}.$$

Finally, divide both sides by $-V$:

$$-\frac{1}{V} \frac{dV}{dy} = \dots = \bar{T} - t.$$

Hence the statement about the bond's *average* life, or effective maturity.

For small movements in the yield, the duration gives a good measure of the change in value with a change in the yield. For larger movements we need to look at higher order terms in the Taylor series expansion of $V(y)$.

One of the most common uses of the duration is in plots of yield versus duration for a variety of instruments. An example is shown in Figure 14.8. Look at the bond marked 'CPU.' This bond has a coupon of 4.75% paid twice per year, callable from June 1998 and maturing in June 2000. We can use this plot to group together instruments with the same or similar durations and make comparisons between their yields. Two bonds having the same duration but with one bond having a higher yield might be suggestive of value for money in the higher-yielding bond, or of credit risk issues. However, such indicators of relative value must be used with care. It is possible for two bonds to have vastly different cashflow profiles yet have the same duration; one may have a maturity of 30 years but an

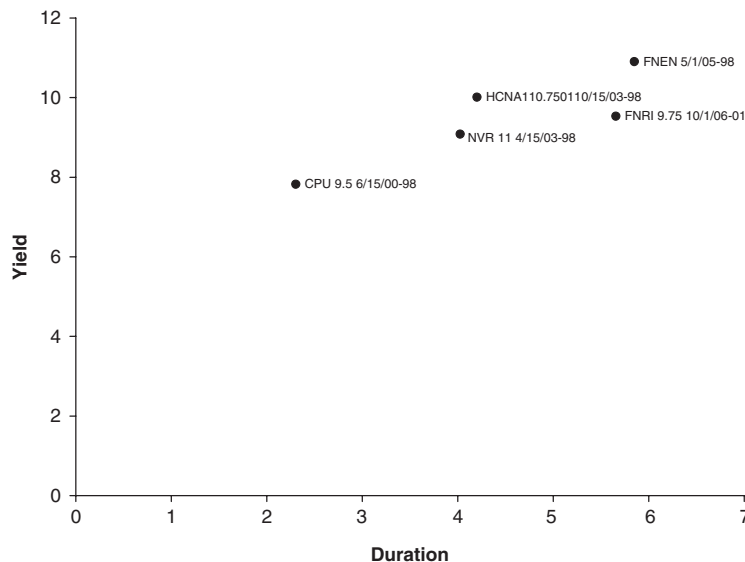


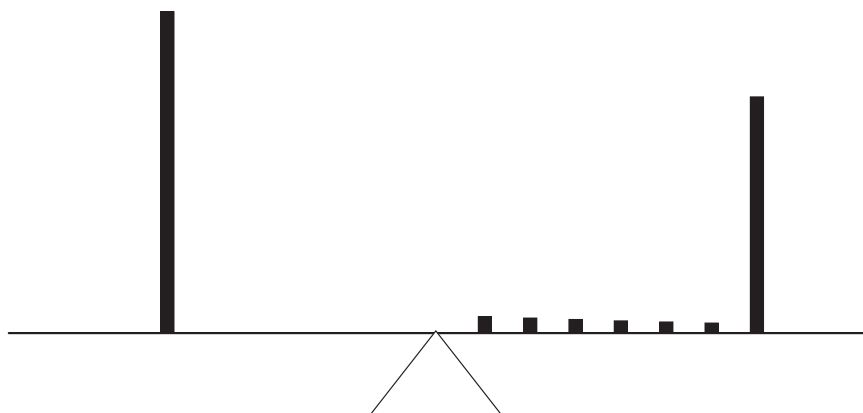
Figure 14.8 Yield versus duration: measuring the relative value of bonds.

average life and hence a duration of seven years, whereas another may be a seven-year zero-coupon bond. Clearly, the former has 23 years more risk than the latter.

Time Out...

Duration = Average life. Why?

The figure below shows a lot of 'cashflows' balancing over a fulcrum. On the right-hand side the arrows are the cashflows associated with the bond we are analyzing. The height of each arrow represents the present value of the cashflow, its distance along the balance from the fulcrum represents the time to the cashflow. The sum of all the heights will therefore represent the known value of the bond. Question, where should an arrow be placed on the left-hand side, having the same height as the sum of all the right-hand arrows, to make a perfect balance? Answer, the distance along from the fulcrum representing the duration of the bond. So, it's rather like an average distance of all the bond's cashflows, or its average life.



14.11 CONVEXITY

The Taylor series expansion of V gives

$$\frac{dV}{V} = \frac{1}{V} \frac{dV}{dy} \delta y + \frac{1}{2V} \frac{d^2V}{dy^2} (\delta y)^2 + \dots,$$

where δy is a change in yield. For very small movements in the yield, the change in the price of a bond



can be measured by the duration. For larger movements we must take account of the curvature in the price/yield relationship.

The **dollar convexity** is defined as

$$\frac{d^2V}{dy^2} = (T - t)^2 P e^{-y(T-t)} + \sum_{i=1}^N C_i (t_i - t)^2 e^{-y(t_i-t)}.$$

and the **convexity** is

$$\frac{1}{V} \frac{d^2V}{dy^2}.$$

To see how these can be used, examine Figure 14.9.

In this figure we see the price/yield relationship for two bonds having the same value and duration when the yield is around 8%, but then they have different convexities. Bond A has a greater convexity than bond B. This figure suggests that bond A is better value than B because a small change in the yields results in a higher value for A. When we develop a consistent theory for pricing bonds when interest rates are stochastic we will see how the absence of arbitrage will lead to relationships between such quantities as yield, duration and convexity, not unlike the Black–Scholes equation.

The calculation of yield to maturity, duration and convexity are shown in the simple spreadsheet in Figure 14.10. Inputs are in the gray boxes.

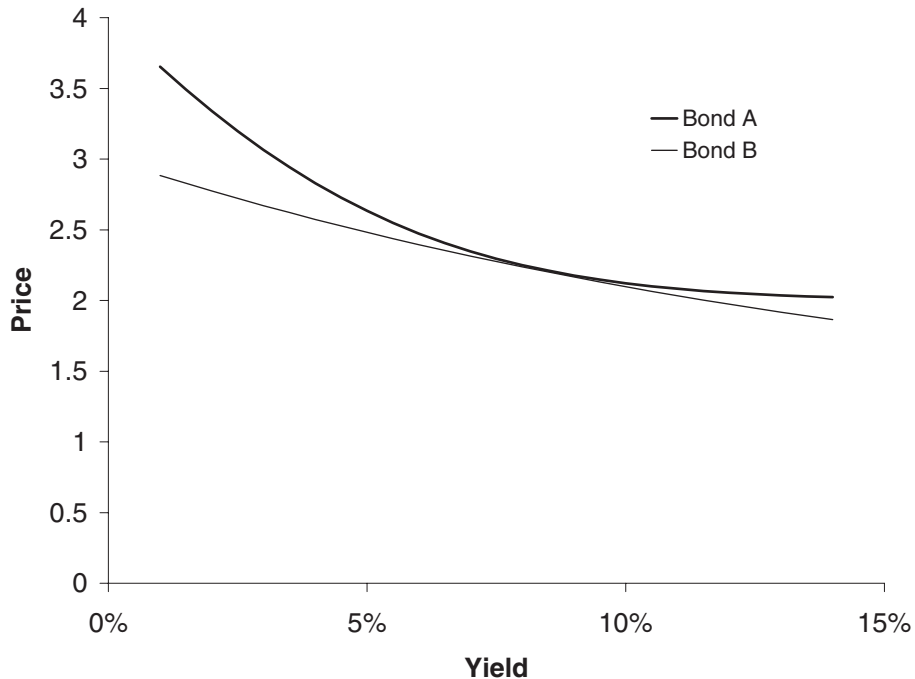


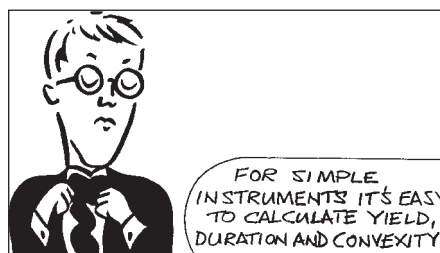
Figure 14.9 Two bonds with the same price and duration but different convexities.

	A	B	C	D	E	F	G	H	I	J	K
1					Date	Coupon	Principal	PVs	Time	Time^2	
2					0				wtd	wtd	
3		YTM	4.95%		0.5	2%		0.0195	0.0098	0.0049	
4		Mkt price	0.921		1	2%		0.0190	0.0190	0.0190	
5		Th. Price	0.921		1.5	2%		0.0186	0.0279	0.0418	
6		Error	1.4E-08		2	2%		0.0181	0.0362	0.0725	
7		Duration	8.2544		2.5	2%		0.0177	0.0442	0.1104	
8		Convexity	76.8728		3	2%		0.0172	0.0517	0.1551	
9					3.5	2%		0.0168	0.0589	0.2060	
10		=SUM(H3:H22)		=C4-C5	4	2%		0.0164	0.0656	0.2625	
11					4.5	2%		0.0160	0.0720	0.3241	
12					5	2%		0.0156	0.0781	0.3903	
13					5.5	2%		0.0152	0.0838	0.4607	
14					6	2%		0.0149	0.0892	0.5349	
15					6.5	2%		0.0145	0.0942	0.6124	
16		=SUM(I3:I22)/C5			7	2%		0.0141	0.0990	0.6929	
17					7.5	2%		0.0138	0.1035	0.7760	
18					8	2%		0.0135	0.1077	0.8613	
19					8.5	2%		0.0131	0.1116	0.9485	
20					9	2%		0.0128	0.1153	1.0374	
21					9.5	2%		0.0125	0.1187	1.1276	
22					10	2%	1	0.6216	6.2161	62.1514	
23											
24		=F20*EXP(-E20*\$C\$3)									
25											
26									=E20*H20		
27											
28											
29											
30											
31											
32											
33											
34											
35											
36											
37											

Figure 14.10 A spreadsheet showing the calculation of yield, duration and convexity.

14.12 AN EXAMPLE

Figure 14.11 shows the yield analysis screen from Bloomberg. The yield, duration and convexity have been calculated for a specific US Treasury. Figures 14.12 and 14.13 show time series of the price and yield respectively.



14.13 HEDGING

In measuring and using yields to maturity, it must be remembered that the yield is the rate of discounting that makes the present value of a bond the same as its market value. A yield is thus identified with each individual instrument. It is perfectly possible for the yield on one instrument to rise while another falls, especially if they have significantly different

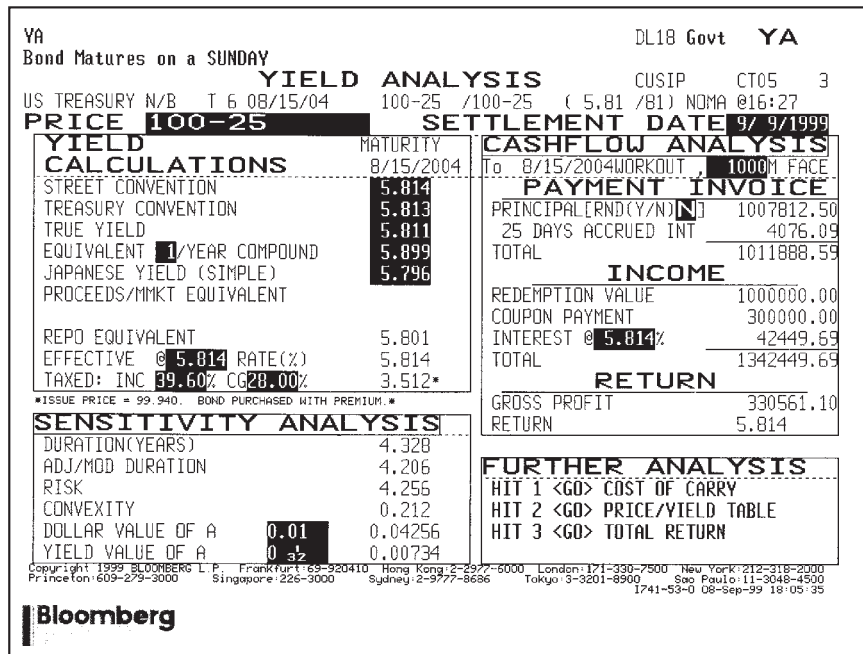


Figure 14.11 Yield analysis. Source: Bloomberg L.P.

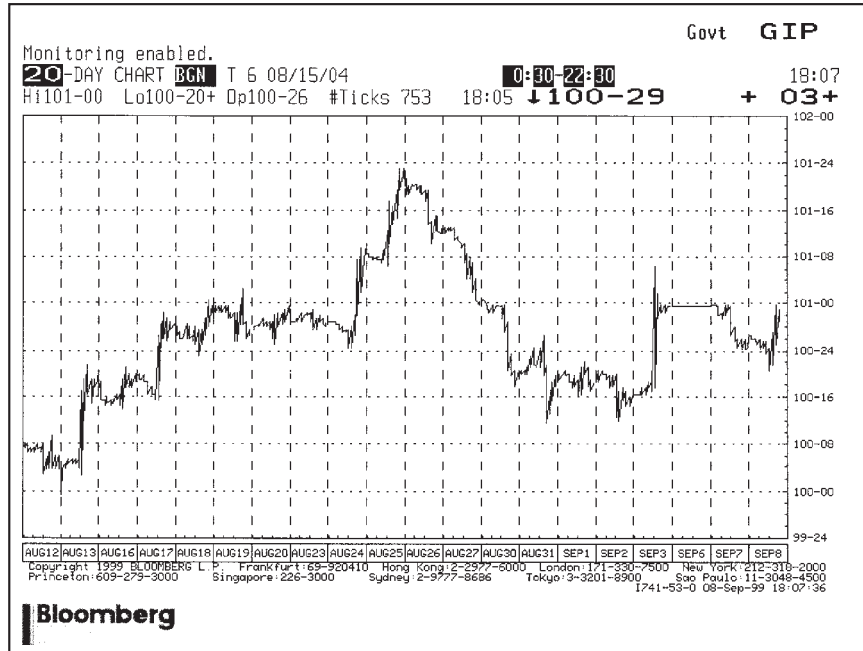


Figure 14.12 Price time series. Source: Bloomberg L.P.

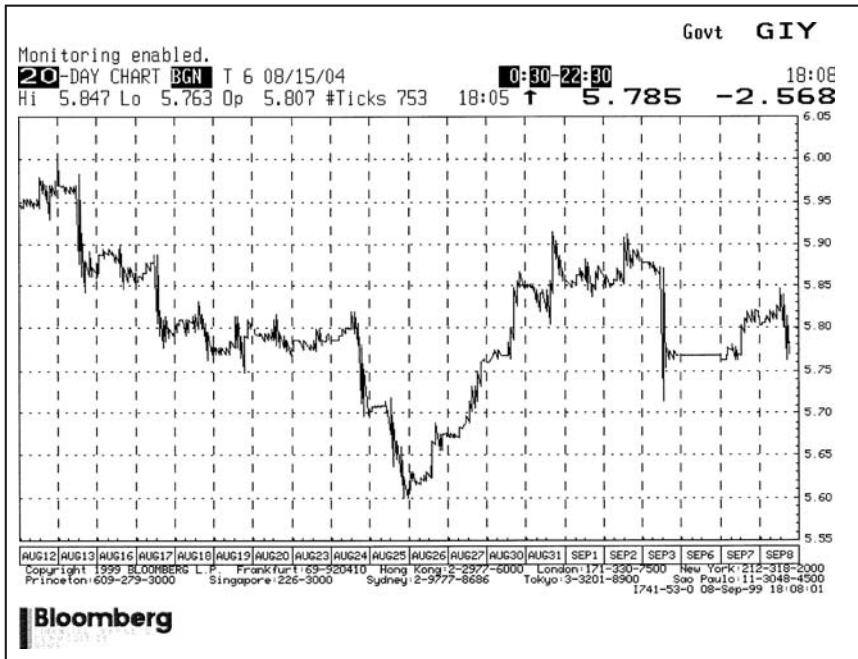


Figure 14.13 Yield time series. Source: Bloomberg L.P.

maturities or durations. Nevertheless, one often wants to hedge movements in one bond with movements in another. This is commonly achieved by making one big assumption about the relative movements of yields on the two bonds. Bond A has a yield of y_A , bond B has a yield of y_B , they have different maturities and durations but we will assume that a move of $x\%$ in A's yield is accompanied by a move of $x\%$ in B's yield. This is the assumption of **parallel shifts** in the yield curve. If this is the case, then if we hold A bonds and B bonds in the inverse ratio of their durations (with one long position and one short) we will be leading-order hedged:

$$\Pi = V_A(y_A) - \Delta V_B(y_B),$$

with the obvious notation for the value and yield of the two bonds. The change in the value of this portfolio is

$$\delta \Pi = \frac{\partial V_A}{\partial y_A} x - \Delta \frac{\partial V_B}{\partial y_B} x + \text{higher-order terms.}$$

Choose

$$\Delta = \frac{\partial V_A}{\partial y_A} / \frac{\partial V_B}{\partial y_B}$$

to eliminate the leading-order risk. The higher-order terms depend on the convexity of the two instruments.

Of course, this is a simplification of the real situation; there may be little relationship between the yields on the two instruments, especially if the cash flows are significantly different. In this case there may be twisting or arching of the yield curve.

14.14 TIME-DEPENDENT INTEREST RATE

In this section we examine bond pricing when we have an interest rate that is a known function of time. The interest rate we consider will be what is known as a **short-term interest rate** or **spot interest rate** $r(t)$. This means that the rate $r(t)$ is to apply at time t : interest is compounded at this rate at each moment in time but *this rate may change*; generally we assume it to be time dependent.

If the spot interest rate $r(t)$ is a known function of time, then the bond price is also a function of time only: $V = V(t)$. (The bond price is, of course, also a function of maturity date T , but I suppress that dependence except when it is important.) We begin with a zero-coupon bond example. Because we receive 1 at time $t = T$ we know that $V(T) = 1$. I now derive an equation for the value of the bond at a time before maturity, $t < T$.

Suppose we hold one bond. The change in the value of that bond in a time-step dt (from t to $t + dt$) is

$$\frac{dV}{dt} dt.$$

Arbitrage considerations again lead us to equate this with the return from a bank deposit receiving interest at a rate $r(t)$:

$$\frac{dV}{dt} = r(t)V.$$

The solution of this equation is

$$V(t; T) = e^{-\int_t^T r(\tau) d\tau}. \quad (14.2)$$

Now let's introduce coupon payments. If during the period t to $t + dt$ we have received a coupon payment of $K(t) dt$, which may be either in the form of continuous or discrete payments or a combination, our holdings including cash change by an amount

$$\left(\frac{dV}{dt} + K(t) \right) dt.$$

Again setting this equal to the risk-free rate $r(t)$ we conclude that

$$\frac{dV}{dt} + K(t) = r(t)V. \quad (14.3)$$

The solution of this ordinary differential equation is easily found to be, dropping the parameter T ,

$$V(t) = e^{-\int_t^T r(\tau) d\tau} \left(1 + \int_t^T K(t') e^{\int_t^{t'} r(\tau) d\tau} dt' \right); \quad (14.4)$$

the arbitrary constant of integration has been chosen to ensure that $V(T) = 1$.

14.15 DISCRETELY PAID COUPONS

Equation (14.4) allows for the payment of a coupon. But what if the coupon is paid discretely, as it is in practice, for example, every six months, say? We can arrive at this result by a financial argument that will be useful later. Since the holder of the bond receives a coupon, call it K_c , at time t_c there must be a jump in the value of the bond across the coupon date. That is, the values before and after this date differ by K_c :

$$V(t_c^-) = V(t_c^+) + K_c.$$

This will be recognized as a jump condition. This time the realized bond price is *not* continuous. After all, there is a discrete payment at the coupon date. This jump condition will still apply when we come to consider stochastic interest rates.

Having built up a simple framework in which interest rates are time dependent I now show how to derive information about these rates from the market prices of bonds.

14.16 FORWARD RATES AND BOOTSTRAPPING

The main problem with the use of yield to maturity as a measure of interest rates is that it is not consistent across instruments. One five-year bond may have a different yield from another five-year bond if they have different coupon structures. It is therefore difficult to say that there is a single interest rate associated with a maturity.



One way of overcoming this problem is to use **forward rates**.

Forward rates are interest rates that are assumed to apply over given periods *in the future* for *all* instruments. This contrasts with yields which are assumed to apply up to maturity, with a different yield for each bond.

Let us suppose that we are in a perfect world in which we have a continuous distribution of zero-coupon bonds with all maturities T . Call the prices of these at time t $Z(t; T)$. Note the use of Z for zero coupon.

The **implied forward rate** is the curve of a time-dependent spot interest rate that is consistent with the market price of instruments. If this rate is $r(\tau)$ at time τ then it satisfies

$$Z(t; T) = e^{-\int_t^T r(\tau) d\tau}.$$

On rearranging and differentiating this gives

$$r(T) = -\frac{\partial}{\partial T}(\log Z(t; T)).$$

This is the forward rate for time T as it stands today, time t . Tomorrow the whole curve (the dependence of r on the future) may change. For that reason we usually denote the forward rate at time t applying at time T in the future as $F(t; T)$ where

$$F(t; T) = -\frac{\partial}{\partial T}(\log Z(t; T)).$$

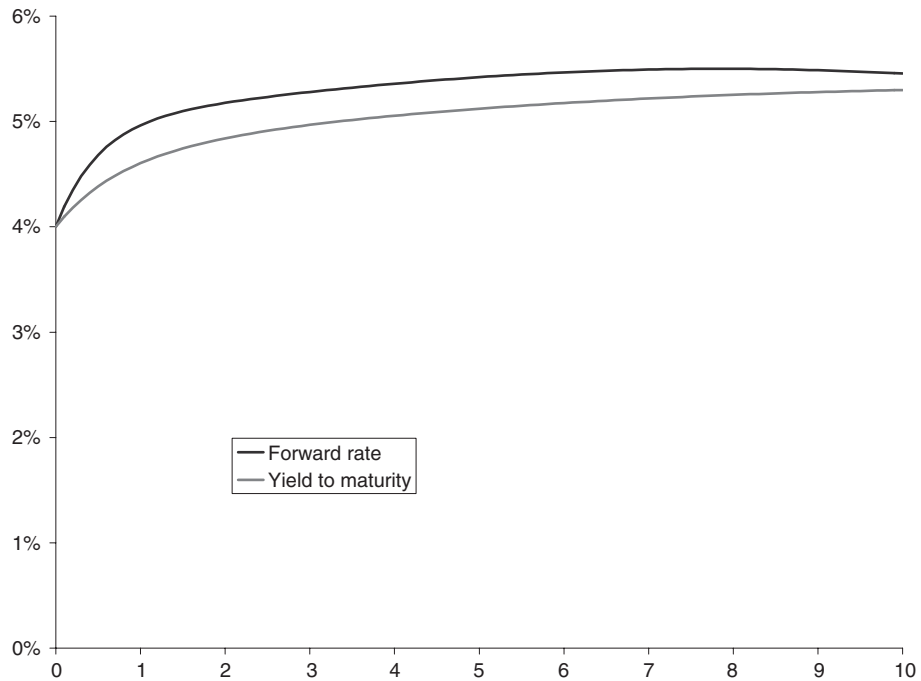


Figure 14.14 The yields and the forward rates.

Writing this in terms of yields $y(t; T)$ we have

$$Z(t; T) = e^{-y(t; T)(T-t)}$$

and so

$$F(t; T) = y(t; T) + \frac{\partial y}{\partial T}.$$

This is the relationship between yields and forward rates when everything is nicely differentiable. See Figure 14.14.

14.16.1 Discrete data

In the less-than-perfect real world we must do with only a discrete set of data points. We continue to assume that we have zero-coupon bonds but now we will only have a discrete set of them. We can still find an implied forward rate curve as follows.

Rank the bonds according to maturity, with the shortest maturity first. The market prices of the bonds will be denoted by Z_i^M where i is the position of the bond in the ranking.

Using only the first bond, ask the question: 'What interest rate is implied by the market price of the bond?' The answer is given by y_1 , the solution of

$$Z_1^M = e^{-y_1(T_1-t)},$$

i.e.

$$y_1 = -\frac{\log(Z_1^M)}{T_1 - t}.$$

This rate will be the rate that we use for discounting between the present and the maturity date T_1 of the first bond. And it will be applied to *all* instruments whenever we want to discount over this period.

Now move on to the second bond having maturity date T_2 . We know the rate to apply between now and time T_1 , but at what interest rate must we discount between dates T_1 and T_2 to match the theoretical and market prices of the second bond? The answer is y_2 which solves the equation

$$Z_2^M = e^{-y_1(T_1 - t)} e^{-y_2(T_2 - T_1)},$$

i.e.

$$y_2 = -\frac{\log(Z_2^M / Z_1^M)}{T_2 - T_1}.$$

By this method of **bootstrapping** we can build up the forward rate curve. Note how the forward rates are applied between two dates, for which period we have assumed they are constant. Figure 14.21 gives an example.

This method can easily be extended to accommodate coupon-bearing bonds. Again rank the bonds by their maturities, but now we have the added complexity that we may only have one market value to represent the sum of several cashflows. Thus one often has to make some assumptions to get the right number of equations for the number of unknowns. See Figures 14.15–14.20.

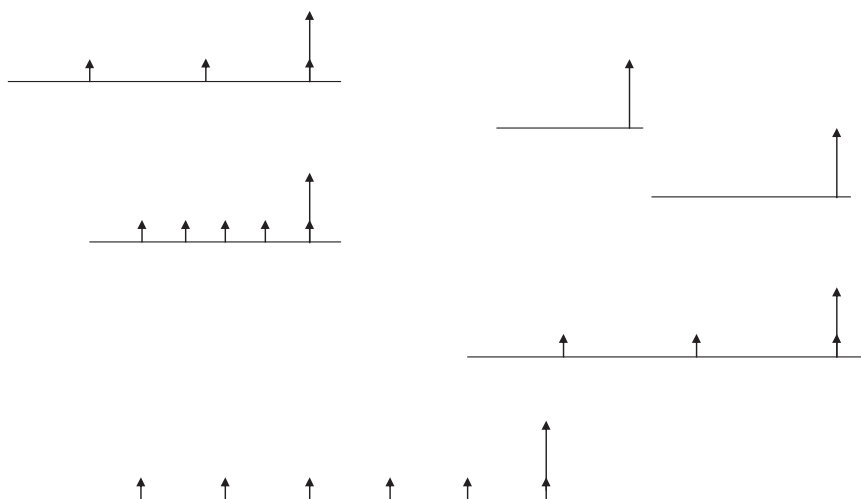


Figure 14.15 The universe of bonds.

14.16.2 On a spreadsheet

Given the market price of zero-coupon bonds it is very easy to calculate yields and forward rates, as shown in the spreadsheet (Figure 14.21). Inputs are in the gray boxes.

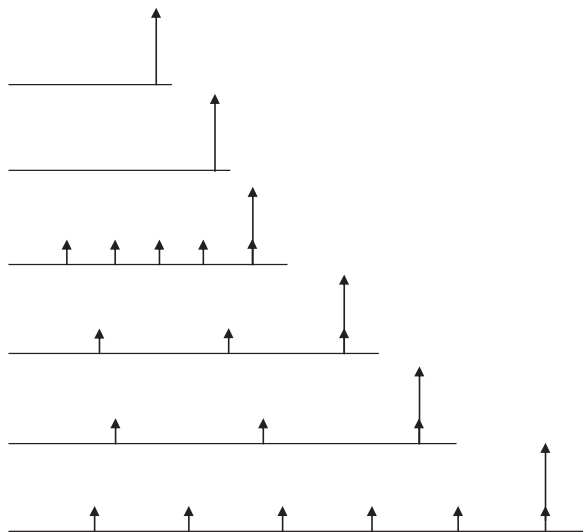


Figure 14.16 The universe of bonds, ranked in order of maturity.

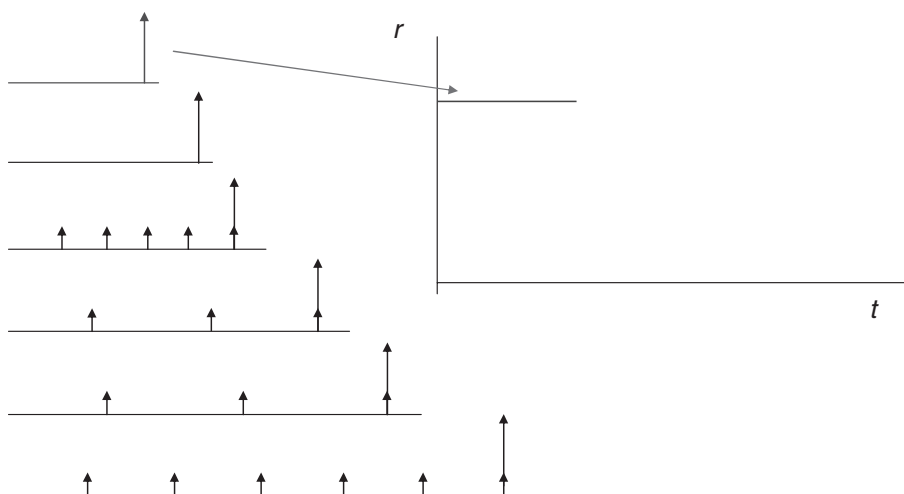


Figure 14.17 The first-maturing bond gives us a forward rate from now until its maturity.

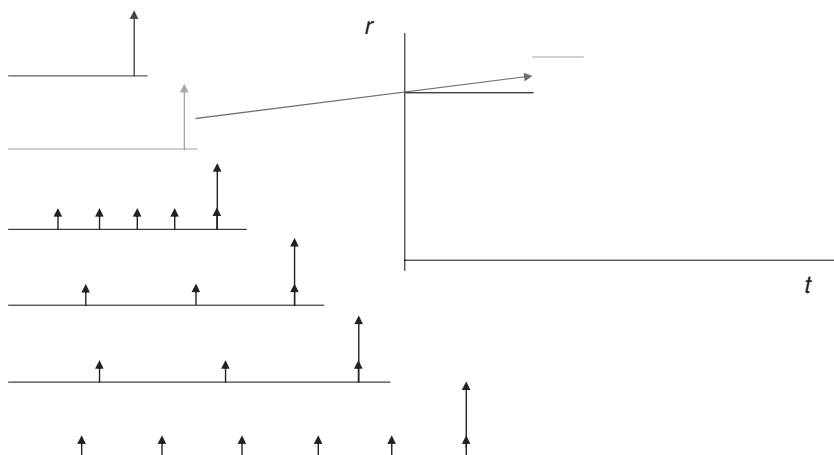


Figure 14.18 The second-maturing bond gives us a forward rate from maturity of the previous bond until its own maturity.

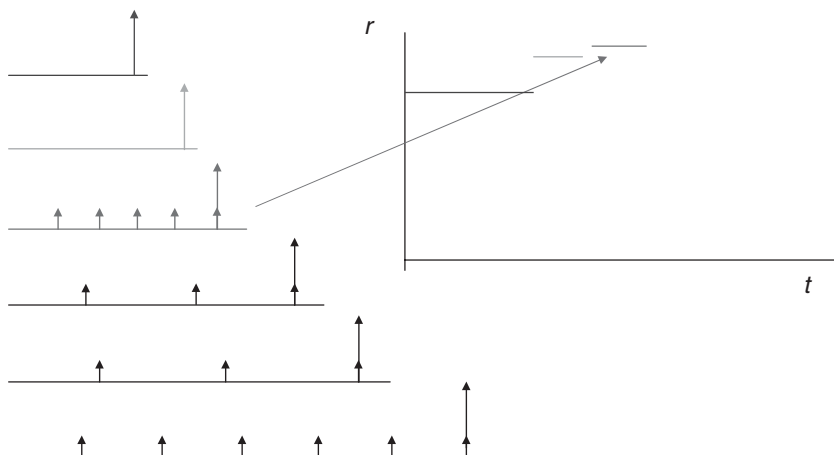
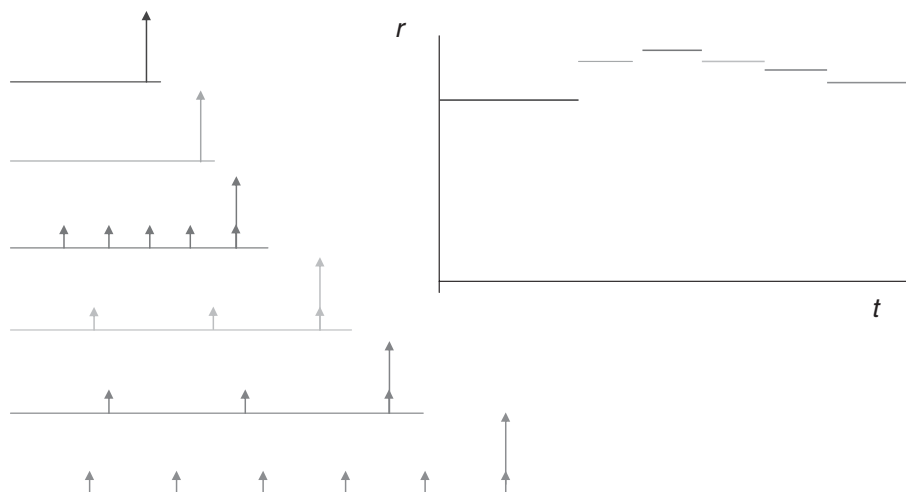
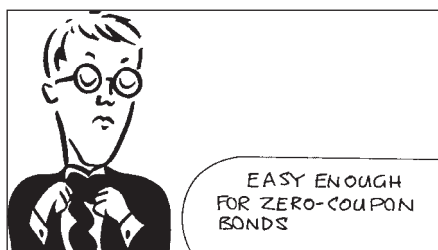


Figure 14.19 The third-maturing bond gives us a forward rate from maturity of the previous bond until its own maturity.

The yields and forward rates for this data are shown in Figures 14.22 and 14.23. Note that in each case the yield begins at zero maturity and extends up to the maturity of each bond. The forward rates pick up where the last forward rates left off.

There are far more swaps of different maturities than there are bonds, so that in practice swaps are used to build up the forward rates by bootstrapping. Fortunately, there is a simple decomposition of swaps prices into the prices of zero-coupon bonds so that bootstrapping is still relatively straightforward. Swaps are discussed in more detail in Chapter 15.

**Figure 14.20** And so on.

	A	B	C	D	E
1	Time to	Market	Yield to	Forward	
2	maturity	price z-c b	maturity	rate	
3	0.25	0.9809	7.71%	7.71%	
4	0.5	0.9612	7.91%	8.12%	
5	1	0.9194	8.40%	8.89%	
6	2	0.8436	8.50%	8.60%	
7	3	0.7772	8.40%	8.20%	
8	5	0.644	8.80%	9.40%	
9	7	0.5288	9.10%	9.85%	
10	10	0.3985	9.20%	9.43%	
11					
12	= -LN(B10)/ A10				
13					
14					
15	= (C10*A10-C9*A9)/ (A10-A9)				
16					

Figure 14.21 A spreadsheet showing the calculation of yields and forward rates from zero-coupon bonds.

14.17 INTERPOLATION

We have explicitly assumed in the previous section that the forward rates are piecewise constant, jumping from one value to the next across the maturity of each bond. Other methods of 'interpolation' are also possible. For example, the forward rate curve could be made continuous, with piecewise constant gradient. Some people like to use cubic splines. The correct way of 'joining the dots' (for there are only a finite number of market prices) has been the subject of much debate. If you want to know what rate to apply to a two-and-a-half-year cashflow and the nearest bonds are at two and three years then you

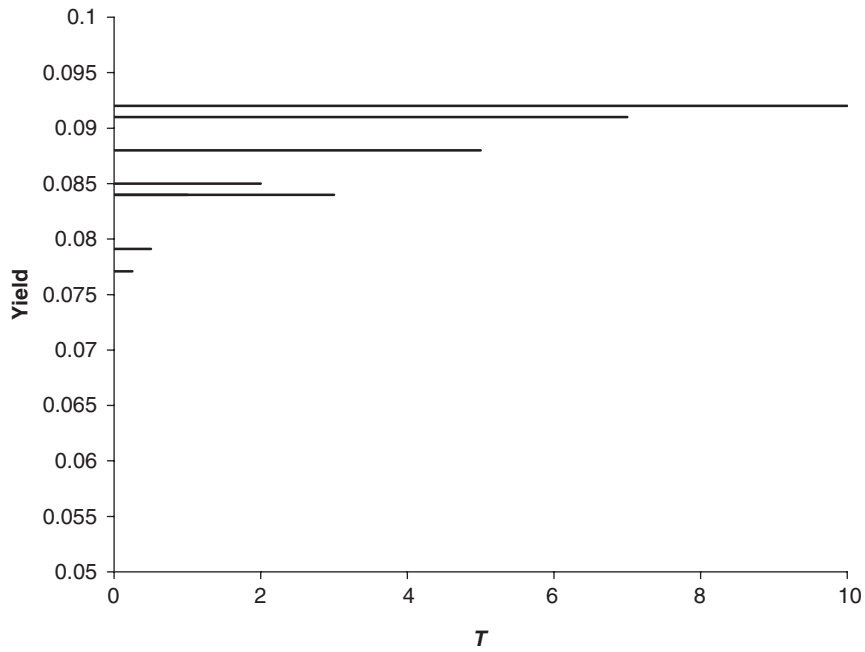


Figure 14.22 Yield to maturities.

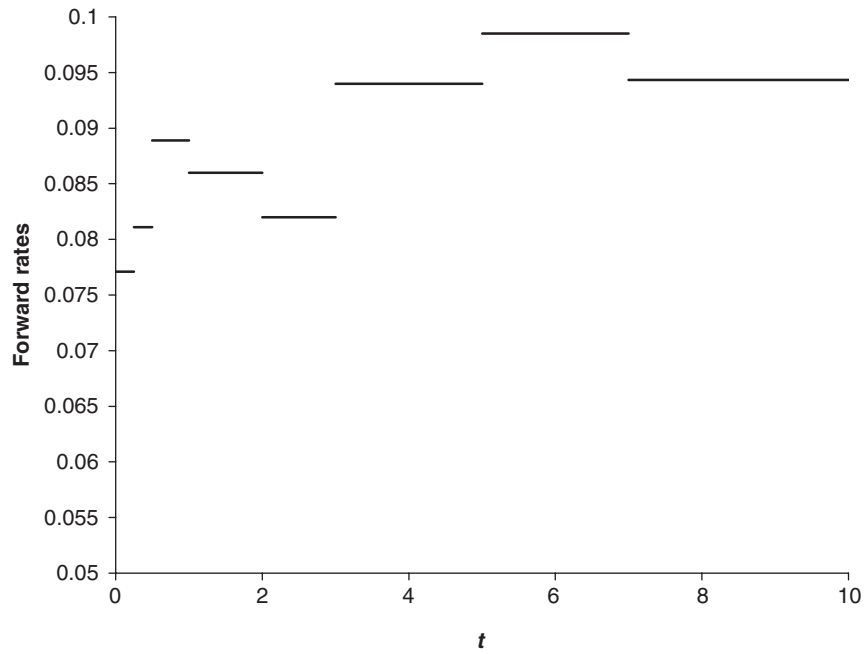


Figure 14.23 Forward rates.

will have to make some assumptions; there is no ‘correct’ value. Perhaps the best that can be done is to bound the rate.

14.18 SUMMARY

There are good and bad points about the interest rate model of this chapter. First, I mention the advantages.

Compare the simplicity of the mathematics in this chapter with that in previous chapters on option pricing. Clearly there is benefit in having models for which the analysis is so simple. Computation of many values and hedging can be performed virtually instantaneously on even slow computers. Moreover, it may be completely unnecessary to have a more complex model. For example, if we want to compare simple cashflows it may be possible to directly value one bond by summing other bonds, if their cashflows can be made to match. Such a situation, although uncommon, is market-independent modeling. Even if exact cashflow matches are not possible, there may be sufficiently close agreement for the differences to be estimated or at least bounded; large errors are easily avoided.

On the other hand, it is common experience that interest rates are unpredictable, random, and for complex products the movement of rates is the most important factor in their pricing. To assume that interest rates follow forward rates would be financial suicide in such cases. Think back to Jensen’s inequality. There is therefore a need for models more closely related to the stochastic models we have seen in earlier chapters.

In this chapter we saw simple yet powerful ways to analyze simple fixed-income contracts. These methods are used very frequently in practice, far more frequently than the complex methods we later discuss for the pricing of interest rate derivatives. The assumptions underlying the techniques, such as deterministic forward rates, are only relevant to simple contracts. As we have seen in the options world, more complex products with non-linear payoffs require a model that incorporates the stochastic nature of variables. Stochastic interest rates will be the subject of later chapters.

FURTHER READING

- The work of Macaulay (1938) on duration wasn’t used much prior to the 1960s, but now it is considered fundamental to fixed-income analysis.
- See Fabozzi (1996) for a discussion of yield, duration and convexity in greater detail. He explains how the ideas are extended to more complicated instruments.
- The argument about how to join the yield curve dots is as meaningless as the argument between the Little-Endians and Big-Endians of Swift (1726).
- See Walsh (2003) for issues concerning curve building.

EXERCISES

1. A coupon bond pays out 3% every year, with a principal of \$1 and a maturity of five years. Decompose the coupon bond into a set of zero-coupon bonds.

2. Construct a spreadsheet to examine how \$1 grows when it is invested at a continuously compounded rate of 7%. Redo the calculation for a discretely compounded rate of 7%, paid once per annum. Which rate is more profitable?
3. A zero-coupon bond has a principal of \$100 and matures in four years. The market price for the bond is \$72. Calculate the yield to maturity, duration and convexity for the bond.
4. A coupon bond pays out 2% every year on a principal of \$100. The bond matures in six years and has a market value of \$92. Calculate the yield to maturity, duration and convexity for the bond.
5. Zero-coupon bonds are available with a principal of \$1 and the following maturities:
 - (a) 1 year (market price \$0.93),
 - (b) 2 years (market price \$0.82),
 - (c) 3 years (market price \$0.74).

Calculate the yield to maturities for the three bonds. Use a bootstrapping method to work out the forward rates that apply between 1–2 years and 2–3 years.

6. What assumption do we make when we duration hedge? Is this a reasonable assumption to make?
7. Solve the equation

$$\frac{dV}{dt} + K(t) = r(t)V,$$

PTE 6 Year Non-Call 2 Year Fixed Rate Step-up Note	
Aggregate Principal Amount	PTE 10,000,000,000
Trade Date	4 November 1997
Issue Date	25 November 1997
Settlement Date	25 November 1997
Maturity Date	25 November 2003
Issue Price	100%
Coupon	Years 1–2: 5.75% Years 3–6: 6.25%
Issuer Optional Redemption	The issuer has the right, but not the obligation, to redeem the Notes at 100% of Nominal, in whole but not in part, on 25 November 1999 with 10 Business Days Prior notice.
Payment Frequency	Annual
Daycount Convention	30/360
Governing Law	English

This indicative termsheet is neither an offer to buy or sell securities or an OTC derivative product which includes options, swaps, forwards and structured notes having similar features to OTC derivative transactions, nor a solicitation to buy or sell securities or an OTC derivative product. The proposal contained in the foregoing is not a complete description of the terms of a particular transaction and is subject to change without limitation.

Figure 14.24 A PTE six-year non-call two-year fixed rate step-up note.

with final data $V(T) = 1$. This is the value of a coupon bond when there is a *known* interest rate, $r(t)$. What must we do if interest rates are not known in advance?

8. Figure 14.24 is a term sheet for a step-up note paying a fixed rate that changes during the life of the contract. Plot the price/yield curve for this product today, ignoring the call feature. What effect will the call feature have on the price of this contract?